# Adversarial Domain Alignment With Contrastive Learning for Hyperspectral Image Classification

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***Abstract*— Recently, deep learning-based hyperspectral image (HSI) classification techniques are flourishing and exhibit good performance, where cross-domain information is usually utilized to reduce the dependence on large labeled samples. However, the gap between the source domain and the target domain makes it difficult to carry out knowledge transfer directly. In this article, an adversarial domain alignment with a contrastive learning method is designed for the HSI classification task to achieve feature consistency that benefits transferring knowledge. In detail, spectral alignment and semantic alignment are conducted at local and global levels, respectively, in an adversarial learning way, and the adversarial loss acts on both source and target domains. In order to learn specific features for objects with different spatial scales, a multiscale selection module is constructed in semantic alignment to select channel features adaptively. Moreover, contrastive learning is employed to increase both robustness and sensitiveness, where augmented data from the same/different samples are forced to be similar/dissimilar with each other. The training process is conducted in a few-shot learning way; then, the few-shot classification loss, the adversarial loss, and the contrastive loss are optimized together. Tested on one source dataset and four target datasets, the experimental results show that the proposed method outperforms the other comparisons.**

***Index Terms*— Adversarial learning, classification, contrastive learning, domain alignment, hyperspectral image (HSI).**

## I. Introduction

HYPERSPECTRAL images (HSIs) contain both 2-D spatial and spectral features, where the spectra consist of hundreds of frequency bands. Recently, HSI has had a wide range of applications in geological exploration [1], agricultural monitoring [2], environment research [3], and so on. As one of the most key techniques in HSI interpretation, HSI classification is to classify HSI pixels into different categories, so as to identify the land covers.

Early on, machine learning methods construct handcrafted features for classification by extracting low-dimensional spectral information [4], [5], [6]. In the past, deep learning has a good performance in image processing; thus, some methods based on deep learning are introduced in HSI classification tasks. Meanwhile, the spatial information and the spectral information of HSI are fed into models together. When more spatial information related to object distribution is considered, it is of great help to further improve the classification accuracy [7], [8]. Chen et al. [9] search for deep features that can achieve good classification results by stacking autoencoders layer by layer. Similarly, Guo et al. [10] propose a deep belief network (DBN) to learn the probability value corresponding to each feature by multiple connected restricted Boltzmann machines (RBMs). It translates HSI classification into how to select deep features of HSI that performs well on classification tasks. Researchers also design the convolutional neural network (CNN) [11] and its variants applied to HSI classification in order to obtain robust features and better classification results. Zhao and Du [12] present a CNN-based spatial feature and a BLDE-based spectral feature fusion method for HSI classification. Roy et al. [13] obtain joint spatial–spectral features and spectral complementary features by 3-D CNN and 2-D CNN, making full use of spatial and spectral information to enhance feature representation. Zheng et al. [14] propose a fast patch-free global learning [field-programmable gate array (FPGA)] method that not only captures spatially global dependent features based on full convolution but also improves computational efficiency by patch-free. Zhu et al. [15] explore the effectiveness of GAN on HSI; the generative-adversarial approach improves the generalization of the model in the case of small samples. Hong et al. [16] propose a new minibatch graph convolutional network (miniGCN) and three ways of fusion with CNN by analyzing CNN and GCN at different levels, and the method introduces the idea of fusion of both into the field of remote sensing. The application of multimodal deep learning (MDL) in remote sensing images compensates for the limitation of a single optical image or synthetic aperture radar (SAR) and improves the accuracy of remote sensing classification by fusing the spectral information of the optical image and the spatial information

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of SAR [17]. Wu et al. [18] propose a new cross-channel reconstruction (CCR) module that implements the information interaction between HSI and SAR by reconstructing features. The low-rank representation network (LRR-Net) displays the HSI anomaly detection method as a deep network at first; in addition, adaptive optimization parameters replace the traditional manual adjustment parameters [19]. Also, some approaches based on attention mechanisms have performed well on classification tasks recently. For example, the attention mechanism in wavelet-attention CNN acts on the wavelet-transformed features [20].

For the second problem, since labeling samples is time-consuming and laborious, it is difficult to obtain a large number of labeled samples by manual methods; thus, small sample tasks are generated. While general supervised learning methods require a large number of labeled samples for network parameter estimation, so many supervised methods have low classification accuracy, some semisupervised and unsupervised methods are proposed for HSI classification tasks. For example, Ling et al. [21] improve classification performance by building semisupervised CNN networks and using unlabeled data. Wang et al. [22] propose a category transfer method that transfers some samples in the unlabeled sample set into labeled samples, so as to improve the classification results by increasing the number of labeled samples. Mei et al. [23] design a joint spatial–spectral network to learn super-pixel features and connect a random forest classifier, which improves the classification performance with unbalanced samples in an unsupervised manner. Romero et al. [24] design an unsupervised network for pixel classification of HSI using a sparse representation of deep convolutional hierarchical features.

Researchers find that the source domain dataset contains a large number of labeled samples, and the knowledge learned from the source domain can be applied to the target domain, thus solving the problem of few labeled samples in the target domain. Therefore, a lot of studies are conducted on how to use cross-domain information to achieve target domain classification. In order to reduce the difference in data distribution between source and target domains, researchers mainly consider two perspectives: data and model. For the former, it can be divided into two types: feature alignment and instance adaptation. Feature alignment generally reduces the distribution difference by aligning the feature space of the source and target domains. Pan et al. [25] directly use spectral feature alignment (SFA) to alleviate inconsistencies in the distribution of source and target domains. Some others transfer discriminative spatial-spectral features to achieve heterogeneous cross-domain learning [26]. Domain adaptation is achieved by calculating domain similarity and domain complexity metrics, and selecting the source domain instance that is most similar to the target domain [27]. The latter fine-tunes the model by loading the model learned in the source domain to the target domain and using a small amount of data to make the learned features perform better for classification on the target domain [28]. To improve the generalization ability of the model in the target domain, Tseng et al. [29] propose a featurewise transformation layer to simulate the feature distribution of different domains. If there are unique classes in the target domain, the source categories in the former cannot provide available knowledge for these classes. The latter can handle the case of category space inconsistency, but the effectiveness of such methods decreases when the domain differences increase to a certain degree.

Therefore, some approaches for severe domain transfer are proposed to solve the cross-domain problem. Li et al. [30] work on reducing the domain discrepancy by an adaptive domain alignment module that improves the classification accuracy in the target domain at the global feature level. Bai et al. [31] propose an adaptive subspace and feature transform-based approach for HSI classification tasks. The feature transform module enhances the ability to generalize the extracted features to the target domain and reduces the domain differences at the feature representation level [31]. In order to reduce the domain differences nearly, Wang et al. [32] introduce a local alignment module to align the global and local features to further improve the accuracy of the results. The method combines 3-D convolution and GAN to extract both spectral and spatial features using the generator's 3-D convolution kernel and the discriminator to find a decision boundary for known and unknown categories in the target domain [33]. The cycle auxiliary classifier generative adversarial network (Cycle-AC-GAN) contains two generators as a shared two-branch mapping, i.e., target-to-source and source-to-target generators, so that the generated samples of the two domains maintain cycle consistency. In addition, each domain contains two classifiers to form a discriminator, which can accomplish the classification task in addition to judging which domain the samples come from [34]. The gradient feature-oriented 3-D domain adaptation (GF-3DDA) method utilizes 3-D Sobel–Gabor gradient (3DSGG) and 3-D derivative-of-Gaussian–Gabor (3DDGG) features extracted from 2-D Sobel gradient (SG) and 2-D derivative-of-Gaussian (DtG) to achieve 3-D domain adaptation [35]. The two-branch attention adversarial domain adaptation (TAADA) network refers to the generative-adversarial network structure, where the two-branch feature extraction (TBFE) module designed in the generator takes into account the spectral–spatial features with attention, and then, the extracted domain-invariant features are mitigated for domain shifting by the discriminator [36]. The coarse-to-fine joint distribution alignment (JDA) framework not only learns a coarse-grained source–target domain consistency representation by learning a high-level domain-independent representation in a shared latent space but also mitigates class-level (fine-grained) by aligning the joint probability distributions of the two domains interdomain feature inconsistency [37]. The graph domain adversarial network with the dual-weighted pseudolabel loss (GDAN-DWPL) method uses a GCN to extract spatial–spectral discriminative features, and in order to reduce the discrepancy in the alignment of class-level probability distributions due to inaccurate pseudolabeling in the target domain, spatial and confidence weights are introduced to further correct for pseudolabel predictions [38]. There is also a way to fuse multiple source domains through an attention mechanism, thus dynamically adjusting the required source domain information to achieve a satisfactory result [39].

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Since only some domain-independent features are extracted for domain transfer using residual networks in [30], more domain-relevant features are nontransferable for target domain classification. In order to retain more complete domain-specific information to improve the classification accuracy of the target domain, we introduce two parameters in the feature mapping process at the shallow level of the network, and each domain can learn its own specific parameters to retain more complete domain-specific features since only the final layer features extracted from the residual network structure are used to achieve data distribution alignment in [32] but ignore the effect of the different scales of the ground features on the extraction results. Therefore, we introduce a multiscale selection (MSS) module and set the convolution kernel with different scales to adaptively learn discriminative features. Feature enhancement is used to further promote the consistent feature representation again. In [37] and [38], the classification results are improved with the help of pseudolabels generated for the target domain samples, but the process is costly and does not fully guarantee the accuracy of the pseudolabels. Therefore, we choose to use a contrastive learning strategy to improve the classification results without introducing additional labels. In this process, each sample is self-supervised, and the feature extractor is optimized by the contrastive loss of all the samples so that the model extracts more discriminative features, which then helps to improve the accuracy of the different domain alignments. Also, the augmented samples increase the robustness of the model in the face of variations.

In this article, we propose an adversarial domain alignment with a contrastive learning method for the task of HSI classification. We combine adversarial learning and contrastive learning for solving cross-domain HSI classification tasks. First, the spectral features of the source and target domains are mapped to learn a shared latent space, and the spectral features are aligned by using adversarial learning. In this process, the parameters preserve domain-specific features, which are nontransferable knowledge to the target domain. In addition, the final layer of the network extracts high-level semantic features that learn multiscale features and further preserve the feature consistency through feature enhancement, and the semantic features are further aligned using adversarial learning. These ensure that the model achieves spectral and semantic level alignment of features at different levels of the network (local and global features), reducing domain differences. Data augmentation is performed for both domains, the augmented samples are passed through the feature extractor to learn discriminative features, which are then aggregated, and the similarity to other features is computed for each feature separately. Contrastive learning learns the ability to classify the categories by self-supervision and optimizes the feature extractor by contrastive learning loss to increase the robustness and sensitivity of the network. The main contributions of this article are summarized as follows.

1) We combine the adversarial learning strategy and the contrastive learning strategy to achieve cross-domain few-shot HSI classification. The method reduces the differences between source and target domains at the spectral and semantic levels in an adversarial learning way and solves the small sample problems to some extent using contrastive learning.
2) In the part of adversarial domain alignment, it consists of both spectral alignment and semantic alignment. The spectral alignment preserves more discriminative features and domain common features. In addition, an MSS module is introduced for semantic alignment, which captures object features at different scales and further enhances the consistent representation of high-level semantic features.
3) To capture an essential representation of source and target data, contrastive learning is employed in a self-supervised way, where numerous unlabeled samples are fully used by data augmentation. It improves both the robustness and sensitiveness of the feature encoder and improves the classification result further. Moreover, it solves the small sample problems to some extent since no additional labels are required.

The remainder of this article is organized as follows. Section II describes some advances in implementing HSI classification similar to this work and the related contents covered in this article. Section III presents the detailed structure of the proposed adversarial domain alignment with a contrastive learning method for HSI classification. Section IV is about the experiment results and analysis of the method. Section V gives a conclusion.

## II. Related Work

### A. Cross-Domain Learning

Cross-domain learning mainly addresses the problem of how to better transfer the knowledge learned in the source domain to the target domain when domain differences arise between the source and target domains. Cross-domain learning mainly consists of domain adaptation methods and fine-tuning methods.

For the domain adaptation approach, it usually reduces the domain gap by aligning the embedding spaces of the source and target domains. Xia et al. [40] propose a rotation-based ensemble and transfer component analysis (TCA) module, by which the induced features are fed into the RF classifier, and it achieves domain adaptation by integrating each transfer component. Fan et al. [41] design an ideal regularized kernel subspace alignment (IRKSA) method that enables source domain knowledge migration by using source labels for subspace alignment on the kernel space of the source and target domains. However, these methods assume that the source and target domains are class-consistent, and thus, the lack of prior knowledge leads to poor performance in the presence of unique classes on the target domain.

For the fine-tuning approach, it first trains a well-supervised network with a large number of labeled samples, i.e., a pretrained model, and then loads the model into the target network and fine-tunes the parameters using a small number of labeled target domain samples. This enhances the applicability of the model on the target domain dataset. Mei et al. [28] construct a new classification CNN (C-CNN) and a feature-learning CNN (FL-CNN) to learn spectral–spatial features. Deng et al. [42]

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propose to minimize the bias of both domains by means of a multikernel domain adaptation classifier and active learning. Jiao et al. [43] design a deeply fully connected (FC) neural network that spatially aligns the source and target domains to reduce the domain discrepancy. The methods of fine-tuning can solve the problem of differences in the source and target domain categories to some extent.

Cross-domain learning also performs well on HSI classification tasks. Domain adaptation broad learning (DABL) is introduced to the HSI classification task, which utilizes the structure of broad learning (BL) to reduce the source and target domain differences using MMD and manifold regularization at both the mapped feature (MF) and the output layer, respectively [44]. In order to achieve migration between heterogeneous datasets, multiscale spectral–spatial features are first extracted in their respective domains using the multiscale convolutional sparse decomposition (MCSD) method, and then, the SPDA model learns a common latent space, which is optimized via a new optimization method to complete the HSI classification task [26]. The TAADA network refers to the generative-adversarial network structure, where the TBFE module designed in the generator takes into account the spectral–spatial features with attention, and then, the extracted domain-invariant features are mitigated for domain shifting by the discriminator [36]. There is also an HSI classification method that performs random weighted sampling on the source domain and GAN in the target domain and utilizes conditional maximum mean discrepancy (CMMD) to simultaneously reduce the distributional bias of the labels and data [45]. The GF-3DDA method utilizes 3DSGG and 3DDGG features extracted from 2-D SG and 2-D DtG to achieve 3-D domain adaptation and combines the outputs of all the ELM classifiers to obtain HSI classification results [35].

### B. Cross-Domain Few-Shot Learning

As an instance of metalearning, few-shot learning (FSL) usually learns using a small number of labeled samples. In general, an episode is utilized as a task during training, and it involves the support set $S$ (labeled sample set) and the query set $Q$ (unlabeled sample set). Each episode takes $K$ and $N$ samples from class $C$ of $S$ and $Q$, respectively, and it is called $C$-way, $K$-shot. $Q$ is responsible for the measurement of the model trained on $S$. However, the FSL method does not perform well when it encounters large differences in categories. Then, cross-domain learning addresses the case where there is a difference between two domains. Therefore, cross-domain learning is introduced into FSL. In cross-domain FSL, the source domain (with a large number of labeled samples) is denoted by $D_s$. The target domain (with a small number of labeled samples) is denoted by $D_t$, which is further divided into $D_t^l$ (labeled samples) and $D_t^u$ (unlabeled samples) according to whether the samples have labels or not. The number of categories in the source and target domains are denoted as $C_s$ and $C_t$, respectively. In general, the data distribution and categories are not necessarily the same, and the number of categories should satisfy $C_s > C_t$.

Recently, studies have been conducted on how cross-domain FSL methods tackle HSI classification issues. Li et al. [30] construct a deep cross-domain FSL (DCFSL) method to solve the domain transfer for HSI classification. The dual-graph cross-domain FSL (DG-CFSL) approach exploits nonlocal intradomain features to achieve domain alignment with graph structures at the feature and distribution levels, respectively [46]. The graph information aggregation cross-domain FSL (Gia-CFSL) approach simultaneously considers intradomain nonlocal relationships, interdomain features, and distributional similarity [47]. The cross-domain self-taught network (CDSTN) aligns the domain information at different network depths using a semisupervised self-taught strategy and achieves classification using pseudolabels obtained by the pseudolabel refinement (PLR) module [48]. The convolutional transformer-based FSL (CTFSL) method selects the feature extractor convolutional transformer (CT) network that takes into account global information and uses the extracted local-global features to achieve domain alignment [49].

### C. Contrastive Learning

Contrastive learning is a type of self-supervised learning (SSL) that uses its own unlabeled data as supervision to learn high-level feature representations, and it can be applied to a variety of downstream tasks [50]. In detail, contrastive learning compares the current sample with the divided positive and negative samples to learn the representation space of features. It is introduced to classification tasks because it can learn how to distance similar samples closer and farther away from dissimilar samples to distinguish different classes to some extent.

The instance discrimination (InstDisc) approach is first proposed, which designs an agent task and a memory bank to store negative samples, and learns a category-distinguishable representation space by comparing each agent task with negative samples [51]. The momentum contrast (MoCo) designs a new momentum encoder and a queue storage method that improves feature consistency and increases the number of stored samples, resulting in a considerable improvement over previous contrastive learning methods [52]. The simple framework for contrastive learning (SimCLR) adds a projector after the feature encoding layer and uses better data augmentation to further improve accuracy on classification tasks [53]. The swap assignment view (SwAV) predicts different views of features from each other, treats matching problems as predicted problems, and distinguishes different classes of features in a clustering manner [54]. The Bootstrap Your Own Latent (BYOL) approach eliminates the negative samples and clustering centers from the contrastive learning process and uses its own features to predict directly [55].

Recently, contrastive learning has also introduced in HSI classification tasks due to its powerful self-supervised characterization capability. Contrastive GCN (ConGCN) combines contrastive learning and GCN to achieve HSI classification by discovering spectral and spatially supervised information between nodes [56]. The contrastive vision transformer (CViT) introduces contrast learning to ViT, using unsupervised and supervised contrastive loss to enhance feature representation in terms of reducing reconstruction

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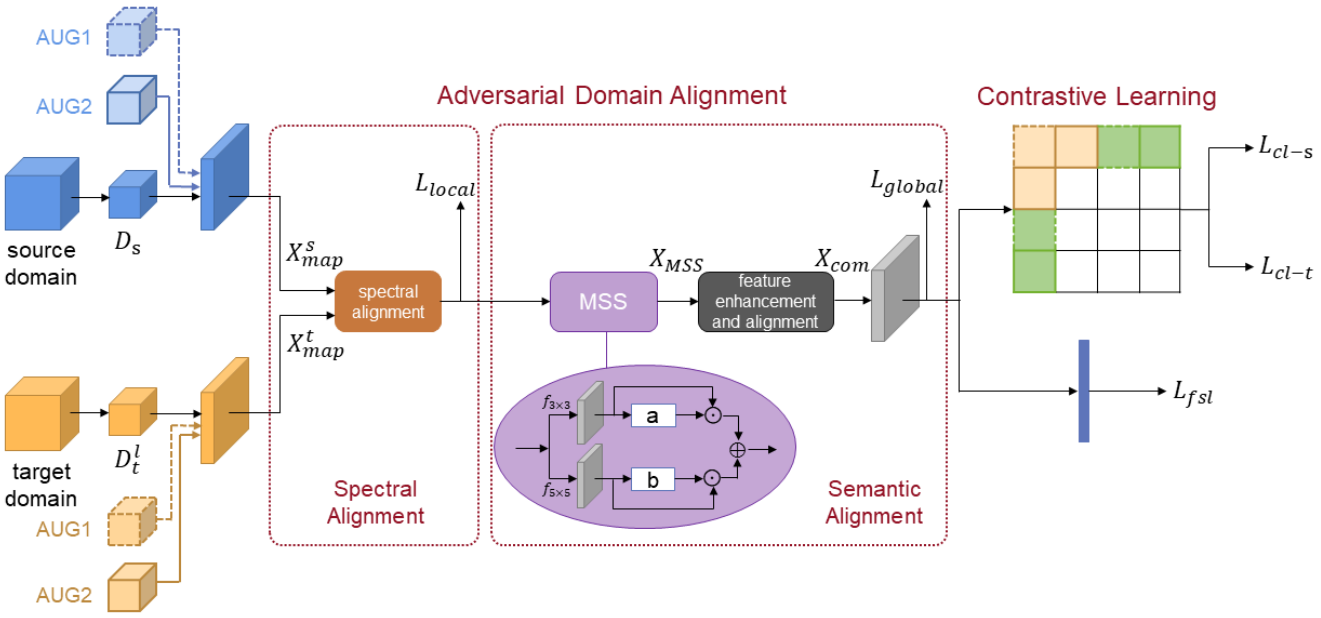



Fig. 1. Framework diagram of the proposed method. The source and target domain datasets are fed into the network together, and domain alignment is achieved by adversarial training to transfer knowledge to the target domain. Among them, spectral alignment and semantic alignment of domain alignment are implemented at the local and global levels, respectively. In addition, contrastive learning enhances the robustness and sensitivity of the whole feature extraction process in a self-supervised manner.

redundancy and enhancing intraclass consistency, respectively [57]. Deep multiview learning (DMVL) combines spectral information from different dimensions of HSI into sample pairs and uses contrastive loss functions to achieve multiview optimization [58]. The refined prototypical contrastive learning network for the FSL (RPCL-FSL) method generates stable support prototypes for FSL classification by constructing contrastive support sets [59]. The augmented HSI sample pairs first train the contrastive SSL model, which is then fine-tuned using labeled samples so that the model extracts reliable features using few labeled samples [60]. Unlike constructing different views of a single sample, the cross-domain contrastive learning (XDCL) method represents shared information in both the spectral and spatial domains by constructing pairs of signals that can represent both domains [61]. The self-supervised contrastive efficient asymmetric dilated network (SC-EADNet) explores multiscale textual features of pixel patches by means of contrastive learning, and efficient asymmetric dilated convolution (EADC) blocks are embedded in contrastive learning architecture, with different dilation rates capturing different scales of features for classification [62]. Contrastive learning is also applicable to unsupervised HSI classification tasks. The spectral–spatial semantic feature learning network (S3FN) aligns the semantic feature space of different views by calculating the contrastive loss, so the whole process achieves classification in an unsupervised manner [63].

The abovementioned contrastive learning methods all achieve classification without introducing additional labels, which provides ideas for solving small sample problems.

## III. Proposed Method

The proposed framework for HSI classification is illustrated in Fig. 1, which implements knowledge transfer across domains by two key components: adversarial domain alignment and contrastive learning. The former reduces the gap between source and target domains in both local and global levels and is achieved by spectral alignment and semantic alignment, respectively. Specifically, the spectral alignment acts on the spectral dimensional features of both domains. With aligned spectral features, the MSS module is constructed to capture multiscale spatial and channel information, and then, the obtained features are enhanced to improve consistency at the global level for semantic alignment. Considering that there are plenty of unlabeled samples and it is worthful to investigate the essential relation among different samples, contrastive learning is employed on the output domain-aligned features from the ADA-encoder, which increases both the robustness and sensitivity of the learned features in a self-supervised manner.

### A. Adversarial Domain Alignment

To obtain consistent features that benefit from transferring both discriminative and common information, an adversarial domain alignment is constructed, which consists of a spectral alignment and a semantic alignment. They are designed to capture consistency at local and global levels, respectively, and trained in an adversarial learning way.

*Spectral Alignment:* In order to reduce the domain gap in the spectral dimension, we propose a spectral alignment module. This module drives the alignment of source and target domains in spectral space.

First, $D_s$ and $D_t^l$ are fed into a mapping network so that the output dimensions are kept consistent. The number of principal components of the PCA operation is 100. The output are denoted by $X_{\text{map}}^s$ and $X_{\text{map}}^t$. Then, the mapping of spectral features is implemented by the spectral alignment module to obtain a shared feature space. The structure of the spectral

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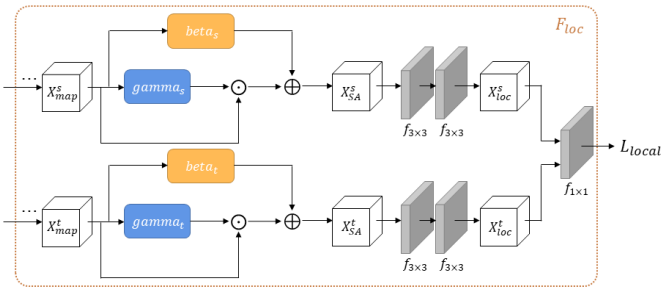



Fig. 2. Structure of the spectral alignment module. Two parameters, gamma and beta, are used to capture features in the spectral dimension of $X^s_{\text{map}}$ and $X^t_{\text{map}}$, respectively.

alignment module is shown in Fig. 2. For each input $X_{\text{map}}$, we obtain its spectral dimension $c$ and then construct two parameters gamma and beta of the same dimensional size to capture of the spectral features. In mathematics, the output $X_{\text{SA}}$ is represented as follows:

$$X_{\text{SA}} = \left(\text{gamma} \odot X_{\text{map}}\right) \oplus \text{beta} \tag{1}$$

where $\odot$ and $\oplus$ denote the operations of elementwise multiplication and summation by corresponding channels, respectively. $X_{\text{SA}}$ denotes features mapped to spectral space after parameter learning. At the beginning, gamma and beta are initialized with all-1 and all-0 vectors, respectively. Therefore, the initialized $X_{\text{SA}}$ is equal to $X_{\text{map}}$.

Second, two convolutional layers are connected, and each layer uses LeakyReLU as an active function to preserve the nonlinear features, which are employed to extract local spectral features from $X_{\text{SA}}$ in the low level. The extracted feature is denoted by $X_{\text{loc}}$ and then passed through a $1 \times 1$ convolutional layer.

These two steps build a spectral feature extractor represented by $F_{\text{loc}}$ and aim at aligning the spectral spaces of the source and target domains. To this end, a discriminator $D_{\text{loc}}$ is employed to distinguish samples from the source or target domain. It generates 1 and 0 for source and target samples, respectively. Specifically, the feature extractor $F_{\text{loc}}$ acts as the generator and captures local features at a low level for spectral space alignment. The corresponding adversarial loss is represented as follows:

$$\begin{aligned}\max_{D_{\text{loc}}} \min_{F_{\text{loc}}} L_{\text{local}} = {} & E_{x_s \sim P_s} \log\left(D_{\text{loc}}(F_{\text{loc}}(x_s))\right) \\ & + E_{x_t \sim P_t} \log\left(1 - D_{\text{loc}}(F_{\text{loc}}(x_t))\right)\end{aligned} \tag{2}$$

where $x_s$ and $x_t$ denote the source and target domain samples, respectively.

With the adversarial training, the learned local features from source and target domains are indistinguishable by discriminator and aligned in spectral space.

*Semantic Alignment:* The spectral alignment maps the two domains into the single spectral space, achieving alignment at the local features level. To further reduce domain differences, we still need to focus on semantic features for alignment, where MSS and feature enhancement and alignment are constructed.

*1) Multiscale Selection Module:* To capture different object information in different domains, we introduce the MSS module. It captures more discriminative and complete features in spatial dimensions by considering different object scales.

The output of spectral alignment module is denoted by $X_{\text{spec}} = F_{\text{loc}}(x)$, where $x$ corresponds to source sample $x_s$ or target sample $x_t$. It is utilized as the input data of MSS, and the structure of the MSS module is illustrated in Fig. 3. The initial extracted features $X_{\text{conf}}$ are obtained by two convolution and one summation operations, as shown in the following equation:

$$X_{\text{conf}} = f_{3\times3}\left(X_{\text{spec}}\right) \oplus f_{5\times5}\left(X_{\text{spec}}\right) \tag{3}$$

where $f_{3\times3}$ and $f_{5\times5}$ denote convolution operations with kernel sizes of $3 \times 3$ and $5 \times 5$, respectively. For convenience, the corresponding outputs are represented by $X^3_{\text{spec}} = f_{3\times3}(X_{\text{spec}})$ and $X^5_{\text{spec}} = f_{5\times5}(X_{\text{spec}})$. The dimension of the obtained $X_{\text{conf}}$ is consistent with $X_{\text{spec}}$.

Next, global average pooling (GAP) is performed for each channel of $X_{\text{conf}}$ to obtain the features of the channel dimension. One FC layer is first utilized to reduce dimension and get the aggregated feature $X_{\text{fc}}$. Then, two FC layers are utilized in parallel to increase the dimension back to the original channel dimension and result in two vectors $A$ and $B$. These two vectors correspond to two kernels $3 \times 3$ and $5 \times 5$ and are used to generate MSS weights $a$ and $b$. In detail, we perform a softmax operation on both with feature weights at the same position summing to 1, as shown in the following equations:

$$a_j = \text{softmax}\left(A_j\right) = \frac{e^{A_j}}{e^{A_j} + e^{B_j}} \tag{4}$$

$$b_j = \text{softmax}\left(B_j\right) = \frac{e^{B_j}}{e^{A_j} + e^{B_j}} \tag{5}$$

where $a_j$, $b_j$, $A_j$, and $B_j$ denote the $j$th ($j = 1, \ldots, c$) elements in $a$, $b$, $A$, and $B$, respectively, and $c$ indicates the number of channels. Notice that $a_j$ and $b_j$ denote the attention of the $j$th channel after softmax processing.

The output of MSS represented by $X_{\text{MSS}}$ is obtained then by (6), where $a$ and $b$ are implicitly spanned to have the same dimension to $X^3_{\text{spec}}$ and $X^5_{\text{spec}}$, respectively,

$$X_{\text{MSS}} = \left(X^3_{\text{spec}} \odot a\right) \oplus \left(X^5_{\text{spec}} \odot b\right). \tag{6}$$

This ensures that features can be learned by adaptively selecting the convolutional kernel with the larger weight, rather than by a global range of fixed convolutional kernels. Moreover, it achieves channel attention in each scale at the same time since different channels are associated with different weights.

*2) Feature Enhancement and Alignment:* To achieve semantic alignment, this part embeds feature alignment similar to (1) and utilizes the residual block structure to get deeper and global features. First, the multiscale feature $X_{\text{MSS}}$ is aligned and enhanced by (7) to increase consistency

$$X'_{\text{SA}} = \left(\text{gamma}' \odot f_{3\times3}(X_{\text{MSS}})\right) \oplus \text{beta}'. \tag{7}$$

Then, a residual structure with three convolutional layers and one skip-connection is conducted, which extracts semantic features at the global level, as shown in the following equation:

$$X_{\text{sem}} = f_{1\times1}\left(X'_{\text{SA}} \oplus f_{3\times3}\left(f_{3\times3}\left(X'_{\text{SA}}\right)\right)\right) \tag{8}$$

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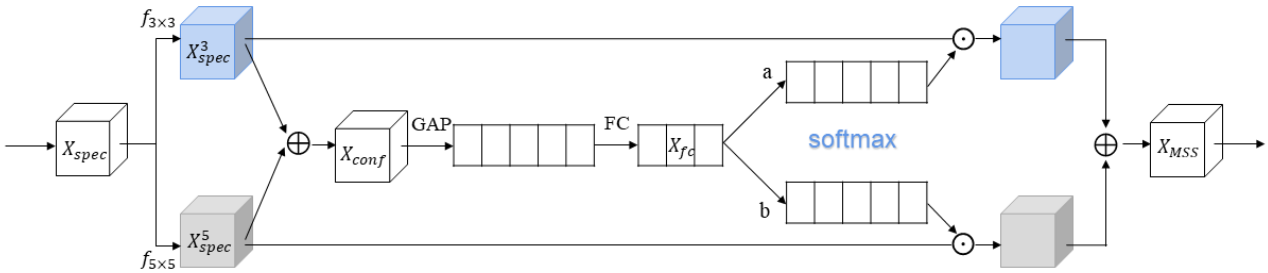



Fig. 3. Structure of the MSS module. Exploring the features of adaptive selection in two scales with a kernel size of 3 × 3 and 5 × 5.

where $f_{1\times1}$ represents convolution operation with 1 × 1 kernel and $X_{\text{sem}}$ is the output of the residual structure and corresponds to the semantic feature in global level.

Semantic alignment is also achieved by adversarial learning similar to that of spectral alignment, where the semantic feature extractor $F_{\text{glo}}$ acts as the generator and the discriminator $D_{\text{glo}}$ is a multilayer perceptron to distinguish source and target domains, as shown in the following equation:

$$\max_{D_{\text{glo}}} \min_{F_{\text{glo}}} L_{\text{global}} = E_{x_s\sim P_s} \log\left(D_{\text{glo}}\left(F_{\text{glo}}(x_s)\right)\right) + E_{x_t\sim P_t} \log\left(1 - D_{\text{glo}}\left(F_{\text{glo}}(x_t)\right)\right). \quad (9)$$

### B. Contrastive Learning

For the task of HSI classification based on cross-domain learning, reducing the gap between source and target domains is convenient for transferring common knowledge between source and target domains, where source samples and target samples are represented in the same feature space. However, it is also important to retain the sensitivity to different samples in a certain domain so that specific knowledge could be reserved and benefit the classification task.

In our work, a contrastive learning module is designed to learn essential and specific features, as illustrated in Fig. 4. Here, two samples from a certain (source or target) domain are represented by $\text{Sample}_1$ and $\text{Sample}_2$. The feature encoder part corresponds to the spectral and semantic alignments, which is represented by $F_{\text{glo}}(\cdot)$, as discussed in (9). The contrastive learning module attends to keep them distinguishable after spectral alignment and semantic alignment. Besides, data augmentation is conducted to generate deformed samples, which helps learn essential representation. In detail, RandomHorizontalFlip and RandomResizedCrop are utilized as two different data augmentation operations and denoted by $\text{AUG}_1$ and $\text{AUG}_2$, respectively. For each sample $\text{Sample}_i (i = 1, 2)$, the augmented one is represented by $\text{Aug}_i^m (m = 1, 2)$, as listed in the following equations:

$$\text{Aug}_i^1 = \text{AUG}_1\left(\text{Sample}_i\right) \quad (10)$$

$$\text{Aug}_i^2 = \text{AUG}_2\left(\text{Sample}_i\right). \quad (11)$$

Augmented data from a certain sample $\text{Sample}_i$ form a positive sample pair $\{\text{Aug}_i^1, \text{Aug}_i^2\}$ and that from different samples form a negative sample pair $\{\text{Aug}_i^m, \text{Aug}_{i'}^{m'}\}$, where $i \neq i'$. With these augmented samples as input data, the output of the global semantic feature extractor is denoted by $F_{\text{glo}}(\text{Aug}_i^1)$ and $F_{\text{glo}}(\text{Aug}_i^2)$. Furthermore, an MLP layer $\text{MLP}(\cdot)$ and a normalization layer $\text{Norm}(\cdot)$ are constructed to generate vector representations of different augmented data, as listed in the following equations:

$$z_i^1 = \text{Norm}\left(\text{MLP}\left(F_{\text{glo}}\left(\text{Aug}_i^1\right)\right)\right) \quad (12)$$

$$z_i^2 = \text{Norm}\left(\text{MLP}\left(F_{\text{glo}}\left(\text{Aug}_i^2\right)\right)\right) \quad (13)$$

where $z_i^1$ and $z_i^2$ are vector representations of augmented data.

Then, for the positive sample pairs, we want the vector representations learned by the model to be similar, while the vector representations between the negative sample pairs are not similar. For each augmented data $z_i^m$, the contrastive loss is constructed based on the infoNCE loss and listed in (14). Notice that $\text{loss}_{\text{pos}}(z_i^m)$ in (15) and $\text{loss}_{\text{neg}}(z_i^m)$ in (16) indicate loss on positive and negative sample pairs, respectively, and $m' \neq m$

$$\min_{F_{\text{glo}}} \text{loss}\left(z_i^m\right) = \text{loss}_{\text{pos}}\left(z_i^m\right) + \text{loss}_{\text{neg}}\left(z_i^m\right) \quad (14)$$

$$\text{loss}_{\text{pos}}\left(z_i^m\right) = -\log \frac{\exp\left(\text{sim}\left(z_i^m, z_i^{m'}\right)/\tau\right)}{\sum_{i'=1}^{B} \exp\left(\text{sim}\left(z_i^m, z_{i'}^1\right)/\tau\right) + \exp\left(\text{sim}\left(z_i^m, z_{i'}^2\right)/\tau\right)} \quad (15)$$

$$\text{loss}_{\text{neg}}\left(z_i^m\right) = \log \frac{\sum_{i'=1, i'\neq i}^{B} \exp\left(\text{sim}\left(z_i^m, z_{i'}^1\right)/\tau\right) + \exp\left(\text{sim}\left(z_i^m, z_{i'}^2\right)/\tau\right)}{\sum_{i'=1}^{B} \exp\left(\text{sim}\left(z_i^m, z_{i'}^1\right)/\tau\right) + \exp\left(\text{sim}\left(z_i^m, z_{i'}^2\right)/\tau\right)} \quad (16)$$

where $\text{sim}(\cdot,\cdot)$ indicates the vector similarity calculation operation, the cosine function is used in our work, $\tau$ is a temperature parameter controlling the attention to negative samples, and $B$ is the number of samples in each episode before augmentation.

In (15), the numerator denotes the similarity value of positive sample pair $\{z_i^m, z_i^{m'}\}$ and the denominator denotes the similarity sum of all sample pairs, while, in (16), the numerator denotes the similarity sum of negative sample pairs $\{z_i^m, z_{i'}^1\}$ and $\{z_i^m, z_{i'}^2\}$, where $i \neq i'$ and all negative sample pairs are taken into consideration. It is noticed that minimizing $\text{loss}_{\text{pos}}(z_i^m)$ makes that positive samples have similar representations and improves the robustness to data deformation, and minimizing $\text{loss}_{\text{neg}}(z_i^m)$ makes that negative samples are distinguishable and increases the sensitiveness to different samples.

The contrastive learning works on both source and target domains, and the final loss is obtained by averaging the losses

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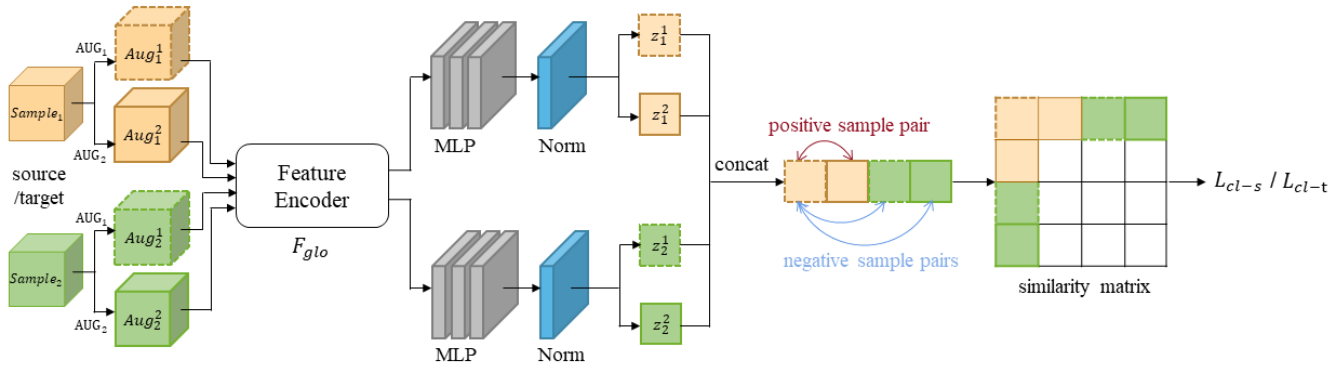



Fig. 4. Contrastive learning procedure. Two samples ($\text{Sample}_1$ and $\text{Sample}_2$) from the source or target domain are taken as an example. Each sample is transformed by two augmented operations $\text{AUG}_1$ and $\text{AUG}_2$ to obtain the augmented samples, which are passed through the feature encoder, the MLP layer, and the normalization layer to obtain the vector representation, respectively. By constructing sample pairs and judging their similarity, the similar samples are close to each other, and dissimilar samples are far away.

on all samples in each episode, as listed in the following equations:

$$L_{\text{cl}-s} = \frac{1}{2B_s}\sum_{i=1}^{B_s}\sum_{m=1}^{2}\text{loss}\left(z_i^m\right) \tag{17}$$

$$L_{\text{cl}-t} = \frac{1}{2B_t}\sum_{i=1}^{B_t}\sum_{m=1}^{2}\text{loss}\left(z_i^m\right) \tag{18}$$

where $B_s/B_t$ indicates the number of samples in each episode for the source/target domain. Thus, summing the losses in the source and target domains produces the total loss $L_{\text{cl}}$ of the contrastive learning module, as listed in the following equation:

$$\min_{F_{\text{glo}}} L_{\text{cl}} = L_{\text{cl}-t} + L_{\text{cl}-s}. \tag{19}$$

For positive sample pairs, the contrastive learning loss guides the feature extractor to learn similar representations and increases the robustness to different data deformations, while, for negative sample pairs, the feature extractor intends to reserve specific information for different samples with the help of contrastive learning, which makes them distinguishable in global semantic representation and benefits the downstream classification task.

Notably, this module makes full use of the large number of unlabeled samples in the target domain, contributing unique and unseen class features of the domain in achieving the classification process. This complements the generalization of the source domain knowledge to further improve the accuracy of classification on the target domain. At the same time, the insufficient number of labeled samples in the target domain is alleviated to some extent.

### C. Training Procedure

We perform FSL in the source and target domains and hope to generalize the knowledge learned in the source domain to the target domain, which helps to improve the feature extraction ability of the target domain in this way. It treats an episode as one task, and the whole training process includes many tasks.

In each episode, $C$ classes are randomly selected from $C_s$ classes, and $K$ labeled samples in each selected class are randomly selected to form the support set, denoted by $S = \{(x_i, y_i)\}_{i=1}^{C\times K}$. Accordingly, in the same $C$ classes, $N$ samples are selected from the remaining samples in each class to form the query set, denoted by $Q = \{(x_i, y_i)\}_{i=1}^{C\times N}$. One training task in each episode consists of a support set and a query set. This division method is used on both $D_s$ and $D_t^l$, where the support/query sets are denoted by $S_s/Q_s$ and $S_t/Q_t$, respectively. Since the number of labeled samples in $D_s$ are much more than that in $D_t^l$, there are much more tasks in $D_s$ than that in $D_t^l$. The probability of class distribution for the query set in the source domain is represented as follows:

$$p(y = k|x \in Q_s) = \frac{\exp\left(-d\left(f_{\text{glo}}(x), c_k\right)\right)}{\sum_{k=1}^{C}\exp\left(-d\left(f_{\text{glo}}(x), c_k\right)\right)} \tag{20}$$

where $c_k$ denotes the category feature of the $k$th class, and $x$ and $y$ are the sample and true labels of the query set $Q_s$, respectively. Moreover, $d(\cdot,\cdot)$ is a distance metric function that calculates the distance between a query sample feature and a certain category feature, and the distance is used as an important reference for classifying the categories. Thus, the FSL loss in source and target domains can be expressed as follows:

$$L_{\text{fsl}-s} = E_{S_s,Q_s}\left[-\sum_{(x,y)\in Q_s}\log p(y = k|x)\right] \tag{21}$$

$$L_{\text{fsl}-t} = E_{S_t,Q_t}\left[-\sum_{(x,y)\in Q_t}\log p(y = k|x)\right]. \tag{22}$$

For the previously mentioned spectral alignment, semantic alignment, and contrastive learning modules, the corresponding losses $L_{\text{local}}$, $L_{\text{global}}$, and $L_{\text{cl}}$ are generated, respectively. Since the source and target domains are trained alternately, the total loss in each training needs to be summed to its FSL loss and other losses for network optimization. The total adversarial optimizations are listed in (23) and (24), where $L_{\text{fsl}}$ is replaced by $L_{\text{fsl}-s}$ or $L_{\text{fsl}-t}$ in the source or target domain

$$\min_{F_{\text{loc}},F_{\text{glo}}} L_{\text{local}} + L_{\text{global}} + L_{\text{cl}} + L_{\text{fsl}} \tag{23}$$

$$\max_{D_{\text{glo}}} L_{\text{local}} + L_{\text{global}}. \tag{24}$$

Thus, labeled samples from the source and target domains are fed into the network together during the training process. The features are aligned at the spectral and semantic levels in an adversarial learning manner. Contrastive learning optimizes the feature extraction capability of the network by identifying

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augmented samples. Finally, the support prototypes and query features of the respective domains are obtained for computing the FSL loss.

In testing, the labeled dataset in the target domain is used to train the KNN classifier, and then, the test set obtains the corresponding features by the feature extractor, after which the label prediction is realized by the classifier.

## IV. Experiment Results and Analysis

### A. Dataset

To verify the effectiveness of the proposed method, we conduct experiments on several HSI datasets. First, the Chikusei dataset is chosen as the source domain, and the other four HSI datasets of the Salinas, the Center of Pavia (PC), the Indian Pines (IP), and the University of Pavia (UP) are chosen as the target domains, respectively. Several datasets are described in turn in the following.

As demonstrated in Fig. 5, the Chikusei dataset describes the Japanese Chikusei HSIs collected by the Hyperspec-VNIR-CIRIS sensor. It contains 2517 × 2335 pixel points, and its spatial resolution is 2.5 m. Moreover, there are 128 bands in the spectral range of 363–1018 nm. Its ground truth contains 19 categories. Table I shows the sample categories and the number of samples included in the Chikusei dataset.

The Salinas dataset describes the Salinas Valley, California imagery collected by the AVIRIS sensor and contains 204 continuous bands after removing some. It has a spatial size of 512 × 217 and a spatial resolution of 3.7 m. It has 16 categories including vegetables, bare soils, vineyard fields, and so on. Fig. 6 shows the pseudocolor image and ground truth. Table II shows the sample categories and the number of samples included in the Salinas dataset.

The PC dataset is captured by the ROSIS sensor with 102 continuous bands for training, a spatial resolution of 1.3 m, and a spatial size of 1096 × 715. It contains nine classes. Fig. 7 shows the pseudocolor image and ground truth. Table III shows the sample classes and the number of samples included in the PC dataset.

The IP dataset is acquired by AVIRIS and has a wavelength range of 400–2500 nm and a spatial resolution of 20 m. The spatial size is 145 × 145. Removing the 20 bands, it has 200 available bands. It has 16 object types, including wheat, corn, woods, and so on. Fig. 8 shows the pseudocolor image and ground truth. Table IV shows the sample categories and the number of samples included in the IP dataset.

The UP is also captured by the ROSIS sensor with 103 continuous bands for training, with a spatial size of 610 × 340 and a spatial resolution of 1.3 m. It contains a number of categories of 9. Fig. 9 shows the pseudocolor image and ground truth. Table V shows the sample categories and the number of samples included in the UP dataset.

### B. Experimental Settings

We design a series of experiments to demonstrate the advancement of our method. In addition, overall accuracy (OA), average accuracy (AA), and kappa are used as evaluation metrics.

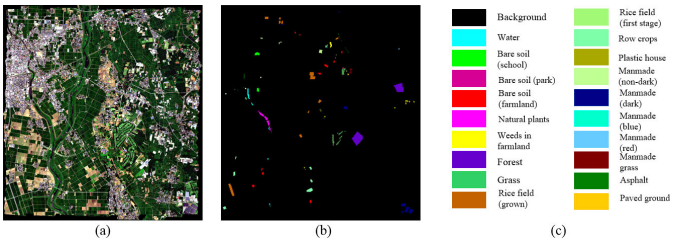



Fig. 5. Chikusei dataset. (a) Pseudocolor image. (b) Ground truth. (c) Sample categories.

TABLE I
Details of the Chikusei Dataset Sample

| Class | Name | Samples |
|---|---|---|
| 1 | Water | 2845 |
| 2 | Bare soil (school) | 2859 |
| 3 | Bare soil (park) | 286 |
| 4 | Bare soil (farmland) | 48525 |
| 5 | Natural plants | 4297 |
| 6 | Weeds in farmland | 1108 |
| 7 | Forest | 20516 |
| 8 | Grass | 6515 |
| 9 | Rice field (grown) | 13369 |
| 10 | Rice field (first stage) | 1268 |
| 11 | Row crops | 5961 |
| 12 | Plastic house | 2193 |
| 13 | Manmade (non-dark) | 1220 |
| 14 | Manmade (dark) | 7664 |
| 15 | Manmade (blue) | 431 |
| 16 | Manmade (red) | 222 |
| 17 | Manmade grass | 1040 |
| 18 | Asphalt | 801 |
| 19 | Paved ground | 145 |

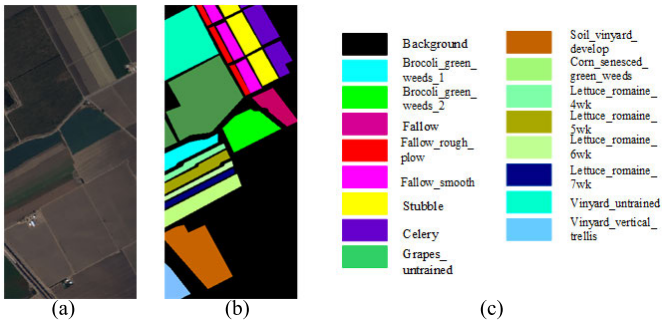


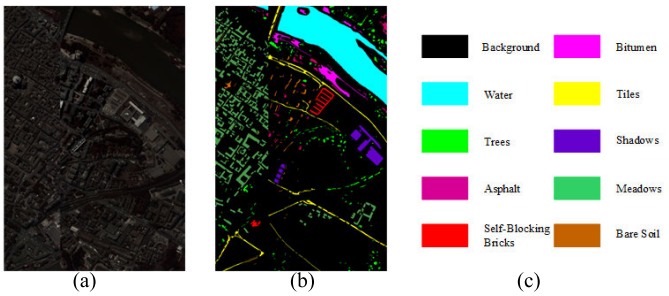

Fig. 6. Salinas dataset. (a) Pseudocolor image. (b) Ground truth. (c) Sample categories.

We use episodes to simulate FSL, and it is necessary to construct episodes in the source and target domains. First, $N_c$ classes are randomly selected from the source domain, and then, $N_k$ samples are selected from each class. The target domain is similar to the source domain. We set $N_k$ to 5 for all four datasets, and $N_c$ is adjusted according to different datasets. For the Salinas, PC, IP, and UP datasets, $N_c$ is 16, 9, 16, and 9, respectively. We also need to set the number of query samples for each class, denoted by $N_q$. To better improve the classification accuracy of the model during testing, $N_q$ is set to the same number of Chikusei dataset categories, i.e., 19. In addition, the network's input is an HSI cube of spatial size 9 × 9. The training episode is 10 000. The SGD is used to speed up the network convergence,

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TABLE II
DETAILS OF THE SALINAS DATASET SAMPLE

| Class | Name | Samples |
|---|---|---|
| 1 | Brocoli_green_weeds_1 | 2009 |
| 2 | Brocoli_green_weeds_2 | 3726 |
| 3 | Fallow | 1976 |
| 4 | Fallow_rough_plow | 1394 |
| 5 | Fallow_smooth | 2678 |
| 6 | Stubble | 3959 |
| 7 | Celery | 3579 |
| 8 | Grapes_untrained | 11271 |
| 9 | Soil_vinyard_develop | 6203 |
| 10 | Corn_senesced_green_weeds | 3278 |
| 11 | Lettuce_romaine_4wk | 1068 |
| 12 | Lettuce_romaine_5wk | 1927 |
| 13 | Lettuce_romaine_6wk | 916 |
| 14 | Lettuce_romaine_7wk | 1070 |
| 15 | Vinyard_untrained | 7268 |
| 16 | Vinyard_vertical_trellis | 1807 |



(a) (b) (c)

Fig. 7. PC dataset. (a) Pseudocolor image. (b) Ground truth. (c) Sample categories.

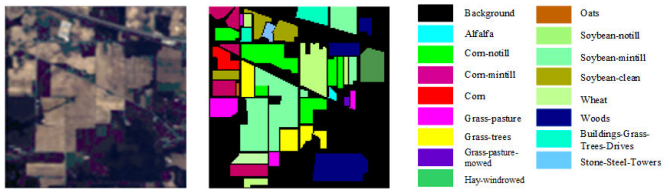
TABLE III
DETAILS OF THE PC DATASET SAMPLE

| Class | Name | Samples |
|---|---|---|
| 1 | Water | 824 |
| 2 | Trees | 820 |
| 3 | Asphalt | 816 |
| 4 | Self-Blocking Bricks | 808 |
| 5 | Bitumen | 808 |
| 6 | Tiles | 1260 |
| 7 | Shadows | 476 |
| 8 | Meadows | 824 |
| 9 | Bare Soil | 820 |

and the learning rate is set to 0.001. In the contrastive learning module, there is a temperature coefficient $\tau$. We look for a suitable value to balance the attention to similar and dissimilar samples, and it has a value of 1.5.

### C. Experimental Results

*1) Comparison Experiments:* We set up a series of comparison experiments to verify the effectiveness of the proposed method, including the traditional method SVM [4], the deep learning method SSRN [64], the FSL methods, DFSL [65] and RN-FSC [66], and the cross-domain learning methods, DCFSL [30], CDSTN [48], GMTN [67], and CTFSL [49]. For the SVM and SSRN methods, five labeled samples from each class of the target domain are taken as the training set, and the other samples as the test set. The DFSL and RN-FSC

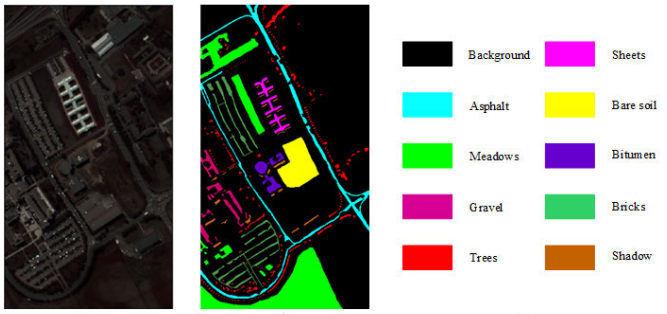


(a) (b) (c)

Fig. 8. IP dataset. (a) Pseudocolor image. (b) Ground truth. (c) Sample categories.

TABLE IV
DETAILS OF THE IP DATASET SAMPLE

| Class | Name | Samples |
|---|---|---|
| 1 | Alfalfa | 46 |
| 2 | Corn-notill | 1428 |
| 3 | Corn-mintill | 830 |
| 4 | Corn | 237 |
| 5 | Grass-pasture | 483 |
| 6 | Grass-trees | 730 |
| 7 | Grass-pasture-mowed | 28 |
| 8 | Hay-windrowed | 478 |
| 9 | Oats | 20 |
| 10 | Soybean-notill | 972 |
| 11 | Soybean-mintill | 2455 |
| 12 | Soybean-clean | 593 |
| 13 | Wheat | 205 |
| 14 | Woods | 1265 |
| 15 | Buildings-Grass-Trees-Drives | 386 |
| 16 | Stone-Steel-Towers | 93 |



(a) (b) (c)

Fig. 9. UP dataset. (a) Pseudocolor image. (b) Ground truth. (c) Sample categories.

TABLE V
DETAILS OF THE UP DATASET SAMPLE

| Class | Name | Samples |
|---|---|---|
| 1 | Asphalt | 6631 |
| 2 | Meadows | 18649 |
| 3 | Gravel | 2099 |
| 4 | Trees | 3064 |
| 5 | Painted metal sheets | 1345 |
| 6 | Bare soil | 5029 |
| 7 | Bitumen | 1330 |
| 8 | Self-Blocking Bricks | 3682 |
| 9 | Shadow | 947 |

methods are FSL methods. The DFSL is usually connected with another classifier to complete the classification task, and here, we choose SVM as the classifier.

For supervised methods SVM and SSRN, only a few samples in the target domain can be used as training set and others as test set so that we ensure class consistency. Five labeled

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TABLE VI
CLASSIFICATION RESULTS OF THE SALINAS DATASET (%)

| Class | SVM [4] | SSRN [64] | DFSL [65] | RN-FSC [66] | DCFSL [30] | CDSTN [48] | GMTN [67] | CTFSL [49] | Proposed |
|---|---|---|---|---|---|---|---|---|---|
| **1** | 97.02 | 99.80 | 77.68 | **100.00** | **100.00** | 97.80 | 97.00 | 99.25 | 98.20 |
| **2** | 88.64 | 99.03 | 95.42 | 97.81 | 97.80 | 98.15 | 98.20 | 98.69 | **99.76** |
| **3** | 80.38 | 98.12 | 96.03 | 94.05 | 95.48 | 88.91 | **99.49** | 98.08 | 91.98 |
| **4** | 95.99 | 91.96 | 99.20 | 70.14 | 99.78 | 99.60 | 98.92 | 99.74 | **99.85** |
| **5** | 92.08 | 97.33 | 83.66 | **98.17** | 96.56 | 94.23 | 86.00 | 94.10 | 95.25 |
| **6** | 97.93 | **100.00** | 90.24 | 99.89 | **100.00** | 97.22 | 99.32 | 99.56 | 99.17 |
| **7** | 95.08 | 99.46 | 98.05 | 83.14 | 98.82 | 94.93 | 99.24 | 99.08 | **99.86** |
| **8** | 64.06 | 70.61 | 80.35 | 87.21 | 75.38 | 87.50 | 84.94 | 83.26 | **90.36** |
| **9** | 97.21 | 96.20 | 95.22 | 84.07 | **100.00** | 97.09 | 96.72 | 98.84 | **100.00** |
| **10** | 70.85 | **93.56** | 78.54 | 88.26 | 90.65 | 85.08 | 89.26 | 89.04 | 90.44 |
| **11** | 93.60 | 96.23 | 94.30 | **99.52** | 99.44 | 91.38 | 93.60 | 98.65 | 98.68 |
| **12** | 96.55 | 95.11 | 97.51 | 76.99 | **99.48** | 96.45 | 96.72 | 99.38 | 99.43 |
| **13** | 95.41 | 96.42 | 96.07 | 49.80 | 98.46 | 96.35 | 90.11 | 99.54 | **100.00** |
| **14** | 93.44 | 91.66 | 98.11 | 78.04 | 98.87 | 98.71 | 86.37 | 99.31 | **100.00** |
| **15** | 62.93 | 75.03 | 70.24 | 73.98 | 76.97 | 77.60 | **83.31** | 80.78 | 73.98 |
| **16** | 83.54 | 96.12 | 88.73 | 88.09 | **100.00** | 87.19 | 98.83 | 91.66 | 99.89 |
| **OA** | 81.03 ±1.08 | 86.21 ±2.53 | 85.63 ±0.63 | 85.19 ±2.14 | 90.56 ±1.69 | 90.89 ±1.45 | 91.80 ±1.06 | 92.20 ±0.52 | **92.80 ±1.22** |
| **AA** | 86.52 ±2.21 | 93.02 ±2.13 | 89.98 ±2.79 | 85.57 ±1.51 | 95.48 ±0.66 | 93.01 ±0.95 | 93.82 ±0.55 | 95.56 ±0.58 | **96.71 ±0.50** |
| **kappa** | 78.77 ±2.34 | 82.65 ±1.28 | 85.04 ±3.98 | 83.58 ±1.05 | 89.52 ±2.01 | 89.83 ±1.61 | 90.88 ±1.15 | 91.33 ±0.58 | **92.65 ±0.43** |

TABLE VII
CLASSIFICATION RESULTS OF THE PC DATASET (%)

| Class | SVM [4] | SSRN [64] | DFSL [65] | RN-FSC [66] | DCFSL [30] | CDSTN [48] | GMTN [67] | CTFSL [49] | Proposed |
|---|---|---|---|---|---|---|---|---|---|
| **1** | 98.65 | **100.00** | 99.05 | 99.26 | 99.45 | 99.79 | 99.87 | 99.54 | 99.77 |
| **2** | 79.27 | 93.66 | 88.24 | 95.07 | 90.49 | 93.10 | **95.61** | 94.66 | 92.84 |
| **3** | 82.62 | 80.48 | 93.40 | 94.20 | 94.49 | 87.86 | 92.50 | 86.09 | **96.53** |
| **4** | 77.41 | 76.52 | 85.14 | 92.85 | **99.66** | 93.51 | 89.60 | 80.30 | 88.36 |
| **5** | 80.06 | 95.31 | 83.76 | 81.63 | 85.54 | 94.33 | 91.28 | 94.47 | **96.10** |
| **6** | 90.13 | 84.24 | 92.28 | 88.46 | 95.06 | 96.50 | 98.49 | 97.44 | **99.55** |
| **7** | 65.75 | **96.92** | 84.96 | 80.93 | 83.14 | 83.87 | 83.25 | 86.32 | 87.37 |
| **8** | 90.18 | 98.30 | 97.01 | 97.88 | 98.66 | 98.81 | 98.32 | 95.49 | **98.84** |
| **9** | 98.56 | 98.02 | 96.20 | 94.23 | 95.14 | 97.78 | **100.00** | 99.97 | **100.00** |
| **OA** | 91.53 ±2.42 | 96.72 ±0.76 | 95.62 ±0.98 | 96.11 ±0.73 | 96.89 ±1.01 | 97.53 ±0.16 | 97.61 ±1.13 | 97.68 ±0.83 | **97.98 ±1.22** |
| **AA** | 85.02 ±3.63 | 91.67 ±0.38 | 90.38 ±2.05 | 92.36 ±0.96 | 93.51 ±0.42 | 93.95 ±0.80 | 94.24 ±0.73 | 93.71 ±0.89 | **95.32 ±0.70** |
| **kappa** | 88.94 ±2.68 | 94.84 ±1.02 | 93.96 ±0.79 | 94.18 ±1.21 | 95.60 ±0.18 | 96.51 ±0.23 | 96.98 ±1.07 | 97.08 ±0.40 | **97.26 ±0.30** |

samples are taken from each class for training. For the FSL methods (DFSL and RN-FSC) and the cross-domain learning method DCFSL, since the target domain requires the transfer of knowledge from the source domain and the categories of the two domains are different, we choose the Chikusei, Houston, Botswana, and Kennedy Space Center (KSC) collection as the source domain. Randomly select 200 and five samples in each category of the source domain and the target domain, respectively. The source domain for the other methods is Chikusei. For the CDSTN, GMTN, and CTFSL methods, five labeled samples from each class in the target domain are taken as training. For our method, five labeled samples from each class are selected to form the support set and 200 samples to form the query set. In addition, both the source and target domains are maintained dimensionally consistent by a PCA operation before feeding into the network. The experiments are executed ten times on each of the four datasets, and the final results are expressed as averages.

Tables VI–IX show the classification results for the comparison experiments on Salinas, PC, IP, and UP, respectively.

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TABLE VIII
CLASSIFICATION RESULTS OF THE IP DATASET (%)

| Class | SVM [4] | SSRN [64] | DFSL [65] | RN-FSC [66] | DCFSL [30] | CDSTN [48] | GMTN [67] | CTFSL [49] | Proposed |
|---|---|---|---|---|---|---|---|---|---|
| **1** | 70.33 | 45.24 | 96.98 | 98.21 | **100.00** | 82.45 | 56.10 | **100.00** | 87.80 |
| **2** | 31.67 | **88.04** | 35.42 | 42.77 | 40.13 | 48.59 | 53.76 | 38.44 | 66.20 |
| **3** | 36.42 | 47.47 | 36.44 | 39.18 | 52.85 | 56.31 | 48.24 | 56.70 | **69.94** |
| **4** | 55.40 | 47.27 | 71.62 | 56.75 | 71.98 | 85.34 | 79.31 | 74.70 | **87.50** |
| **5** | 63.71 | 84.23 | 72.06 | 65.83 | 65.90 | 74.33 | 78.45 | **85.33** | 72.18 |
| **6** | 67.54 | 93.60 | 88.53 | 71.56 | 91.31 | 75.40 | 77.93 | 91.27 | **99.59** |
| **7** | 85.21 | 48.75 | 97.73 | **100.00** | **100.00** | 91.58 | **100.00** | **100.00** | **100.00** |
| **8** | 70.60 | 91.38 | 85.86 | 80.09 | 95.56 | 94.05 | 84.14 | 85.90 | **98.52** |
| **9** | 88.22 | 23.68 | 76.42 | 99.82 | **100.00** | 85.33 | 93.33 | **100.00** | **100.00** |
| **10** | 39.86 | 40.80 | 54.49 | 26.65 | 65.87 | 66.18 | 31.85 | 68.63 | **92.45** |
| **11** | 35.18 | 73.22 | 58.64 | 66.20 | 62.41 | 69.15 | **75.95** | 74.44 | 62.65 |
| **12** | 29.72 | 55.68 | 32.92 | 25.36 | 39.97 | 62.38 | **64.96** | 43.91 | 50.00 |
| **13** | 87.45 | 67.63 | 97.06 | 97.54 | 99.00 | 90.00 | 99.00 | 99.70 | **100.00** |
| **14** | 65.34 | **95.92** | 85.20 | 75.93 | 88.89 | 81.43 | 89.44 | 89.35 | 94.37 |
| **15** | 25.38 | 62.54 | 72.08 | 64.93 | 53.28 | 75.61 | 65.88 | **79.74** | 71.92 |
| **16** | 91.64 | 60.46 | 99.79 | **100.00** | 98.86 | 84.06 | 86.36 | 95.00 | **100.00** |
| **OA** | 42.74 ±3.28 | 62.76 ±0.35 | 61.34 ±1.88 | 57.38 ±1.43 | 65.80 ±2.76 | 69.19 ±4.85 | 68.36 ±1.64 | 70.82 ±1.30 | **76.73 ±1.40** |
| **AA** | 58.20 ±1.49 | 64.12 ±0.03 | 72.48 ±1.55 | 68.21 ±1.17 | 76.63 ±1.60 | 76.39 ±2.10 | 75.14 ±0.32 | 80.19 ±1.60 | **84.57 ±0.54** |
| **kappa** | 39.98 ±2.00 | 58.45 ±0.21 | 55.82 ±0.79 | 52.30 ±0.10 | 61.46 ±1.58 | 65.28 ±4.44 | 64.03 ±1.72 | 66.85 ±1.47 | **73.78 ±1.44** |

TABLE IX
CLASSIFICATION RESULTS OF THE UP DATASET (%)

| Class | SVM [4] | SSRN [64] | DFSL [65] | RN-FSC [66] | DCFSL [30] | CDSTN [48] | GMTN [67] | CTFSL [49] | Proposed |
|---|---|---|---|---|---|---|---|---|---|
| **1** | 86.42 | 91.21 | 71.95 | 66.71 | 80.61 | 78.96 | 76.99 | **92.74** | 87.57 |
| **2** | 82.06 | **93.94** | 89.70 | 93.23 | 87.99 | 91.54 | **96.65** | 83.04 | 86.19 |
| **3** | 38.74 | 55.30 | 44.38 | 48.44 | 56.83 | 59.36 | 55.16 | 53.46 | **93.17** |
| **4** | 66.81 | **95.21** | 83.22 | 92.84 | 94.90 | 89.49 | 70.32 | 94.48 | 83.56 |
| **5** | 94.31 | 99.46 | 98.82 | 98.06 | 97.99 | 96.43 | 98.88 | 99.04 | **99.62** |
| **6** | 39.94 | 45.06 | 82.14 | 59.52 | 83.52 | 71.87 | 76.89 | 80.24 | **86.57** |
| **7** | 37.58 | 77.42 | 60.94 | 69.27 | 74.94 | 79.50 | 78.26 | 80.48 | **89.66** |
| **8** | 70.46 | 73.58 | 72.31 | 68.90 | 76.69 | 75.34 | 74.15 | **90.75** | 72.80 |
| **9** | 99.25 | 95.44 | 91.47 | 99.66 | **99.89** | 90.69 | 98.73 | 99.37 | 98.09 |
| **OA** | 62.57 ±2.28 | 75.82 ±5.86 | 79.19 ±1.11 | 81.92 ±1.43 | 84.49 ±1.18 | 83.67 ±3.45 | 84.97 ±1.95 | 85.03 ±2.02 | **86.33 ±1.45** |
| **AA** | 67.19 ± 1.22 | 80.08 ±2.54 | 75.82 ±1.23 | 78.36 ±0.15 | 83.71 ±1.77 | 81.47 ±3.23 | 82.06 ±1.80 | 85.96 ±1.54 | **88.56 ±0.21** |
| **kappa** | 53.24 ±4.71 | 69.32 ±6.05 | 73.10 ±1.52 | 76.59 ±1.33 | 79.81 ±1.60 | 78.76 ±4.43 | 79.77 ±2.18 | 80.64 ±2.39 | **81.99 ±1.43** |

We list the results for each class, as well as the OA, AA, and kappa values. The best results in each row of the table are bolded. On the Salinas dataset, it can be seen that our method has higher OA, AA, and kappa values than the other methods. Compared with the DCFSL method, our method improves in OA, AA, and kappa by 2.24%, 1.23%, and 3.13%, respectively. Compared with the current more advanced CTFSL method, the proposed method improves the OA, AA, and kappa on 0.6%, 1.15%, and 1.32%, respectively. Certainly, there are improvements over other machine learning, deep learning, and FSL methods. On the PC dataset, our method improves 0.30%, 1.61%, and 0.18% over the best CTFSL method on OA, AA, and kappa, respectively. It can be seen that our method performs better than other comparative methods. On the IP dataset, the CTFSL method is lower than ours on OA, AA, and kappa by 5.91%, 4.38%, and 6.93%, respectively. The other methods have lower values on the three indicators than the proposed method. Moreover, we find better performance on the IP dataset than the other datasets, as evidenced by more gain. Then, on the UP dataset, OA, AA, and kappa are

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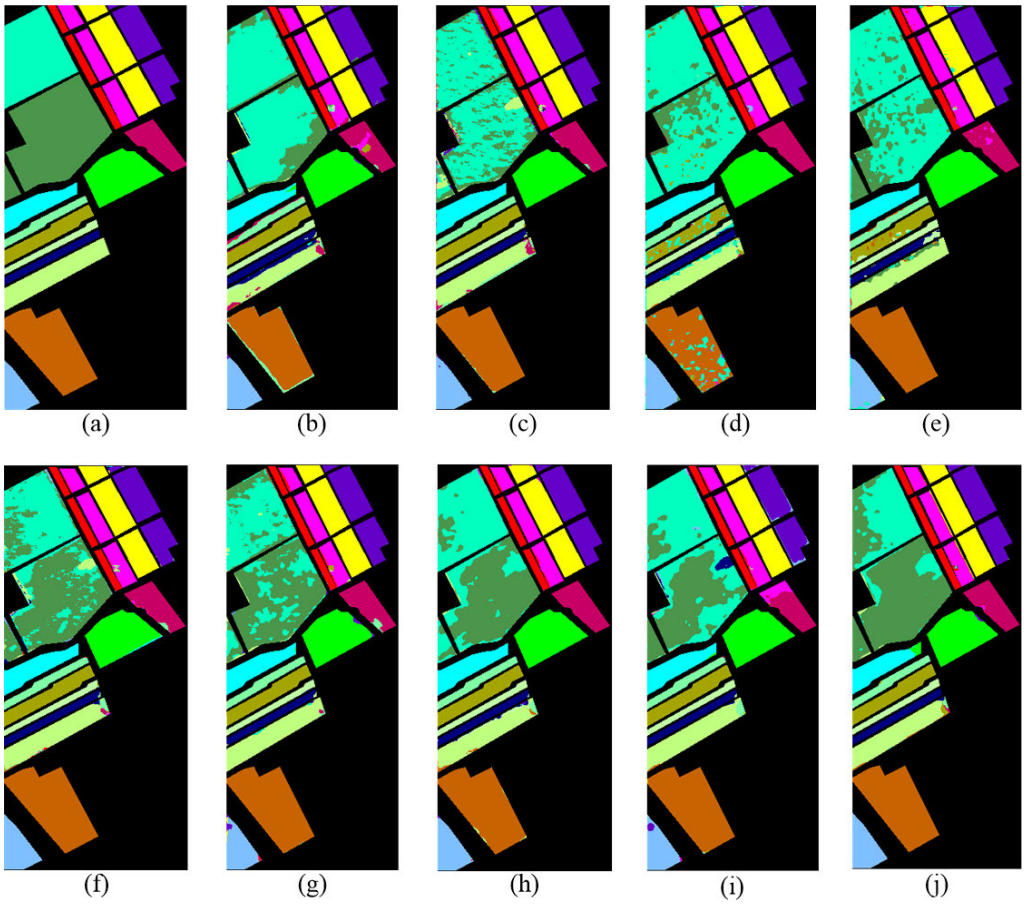



Fig. 10. Classification results for different methods on the Salinas dataset. (a) Ground truth. (b) SVM. (c) SSRN. (d) DFSL. (e) RN-FSC. (f) DCFSL. (g) CDSTN. (h) GMTN. (i) CTFSL. (j) Proposed.

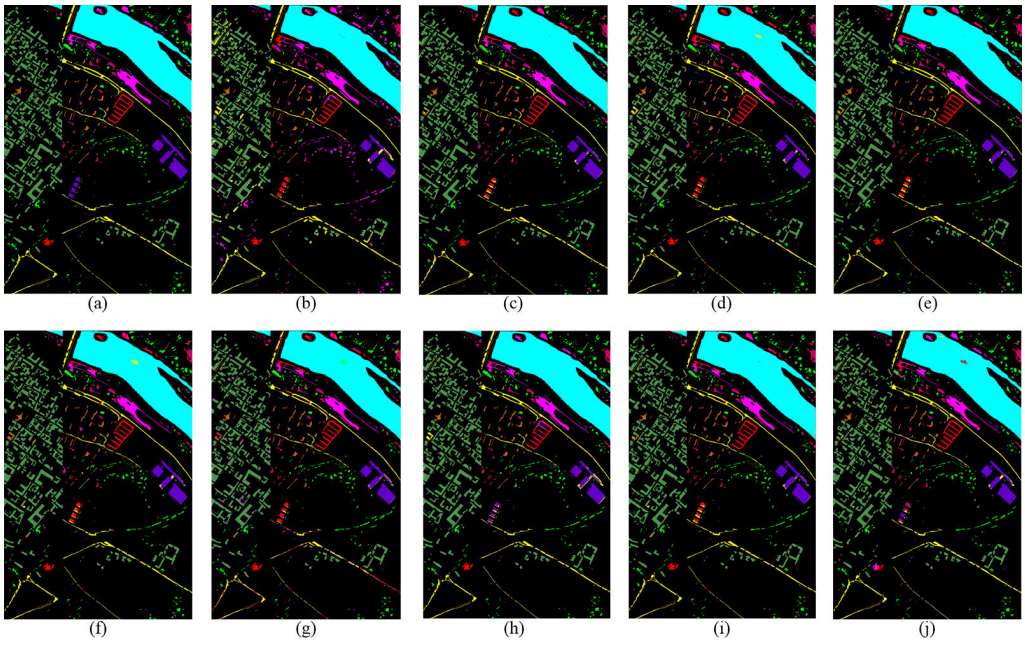



Fig. 11. Classification results for different methods on the PC dataset. (a) Ground truth. (b) SVM. (c) SSRN. (d) DFSL. (e) RN-FSC. (f) DCFSL. (g) CDSTN. (h) GMTN. (i) CTFSL. (j) Proposed.

higher than the best by 1.30%, 2.60%, and 1.35%, respectively. Figs. 10–13 show the comparison results from the view of the results visualization. It can be seen more clearly that our method is more accurate than the other methods on the four datasets. Fig. 14 shows the effect of the spectral alignment module and the semantic alignment module on the feature distribution on the IP dataset. Different colors indicate different class features. It can be seen that both modules

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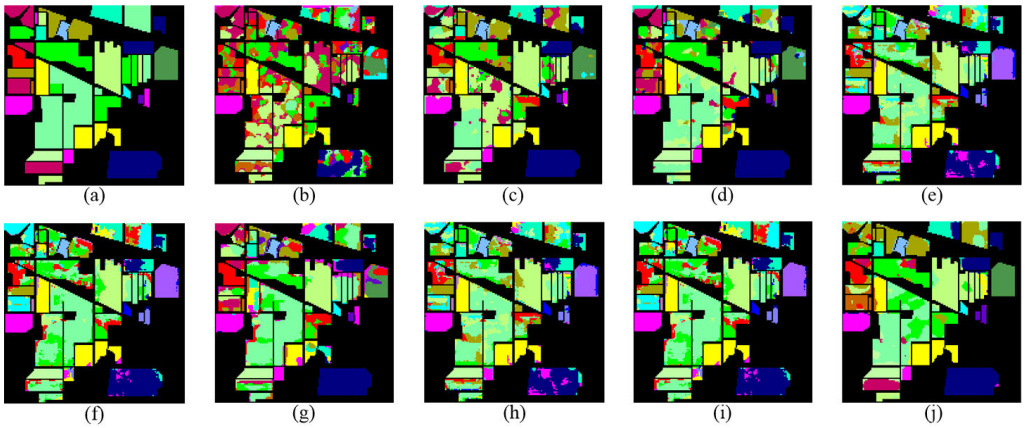



Fig. 12. Classification results for different methods on the IP dataset. (a) Ground truth. (b) SVM. (c) SSRN. (d) DFSL. (e) RN-FSC. (f) DCFSL. (g) CDSTN. (h) GMTN. (i) CTFSL. (j) Proposed.

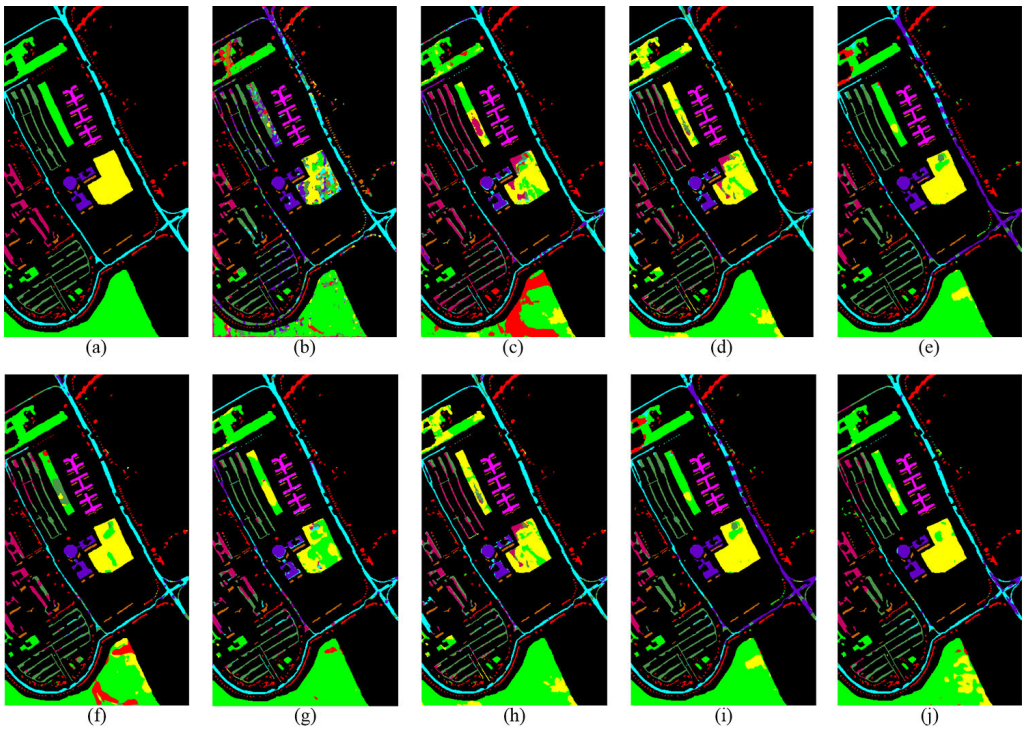



Fig. 13. Classification results for different methods on the UP dataset. (a) Ground truth. (b) SVM. (c) SSRN. (d) DFSL. (e) RN-FSC. (f) DCFSL. (g) CDSTN. (h) GMTN. (i) CTFSL. (j) Proposed.

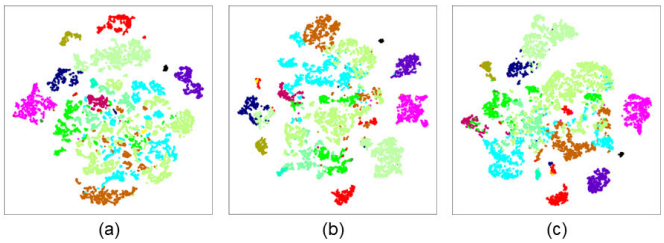



Fig. 14. Feature visualization results of the t-SNE algorithm on the IP dataset. (a) Features obtained by feature extractor without spectral alignment module and semantic alignment module. (b) Features obtained by feature extractor adding spectral alignment module. (c) Features obtained by feature extractor adding semantic alignment module.

enable aggregation of intraclass features and further distances of interclass features, i.e., clearer margins between different classes to some extent.

Based on the analysis, we obtain the relationship between the different methods and the results. The SVM method maps the samples into the feature space and achieves the classification of the features by a linear classifier. The SSRN method uses the 3-D CNN as a feature extractor, and the jointly acquired spatial–spectral features help the classification task to produce good results. In contrast to SSRN, SVM focuses on the spectral features of HSI rather than the joint spatial–spectral features. Therefore, this proves that considering both spatial and spectral information is beneficial to improve classification accuracy. For the DFSL and the RN-FSC methods, they utilize the idea of FSL to achieve classification, i.e., applying metaknowledge learned from the source domain with a large number of labeled samples to the target domain with few labeled samples. The experimental results show that FSL can solve the problem of few samples well. The RN-FSC method analyzes whether the sample features match or not by calculating their distances, which can be regarded as providing a nonlinear classifier for judging relationships. It can also be seen that the nonlinear

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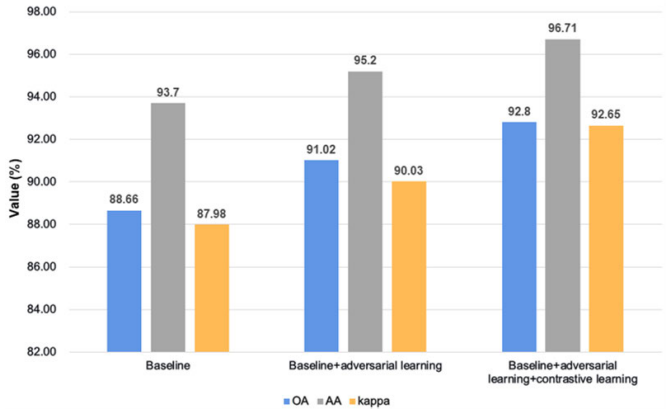



Fig. 15. Ablation experiment results on the Salinas dataset.

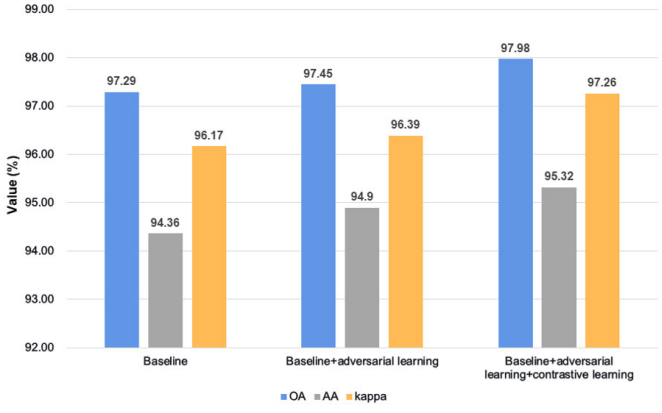



Fig. 16. Ablation experiment results on the PC dataset.

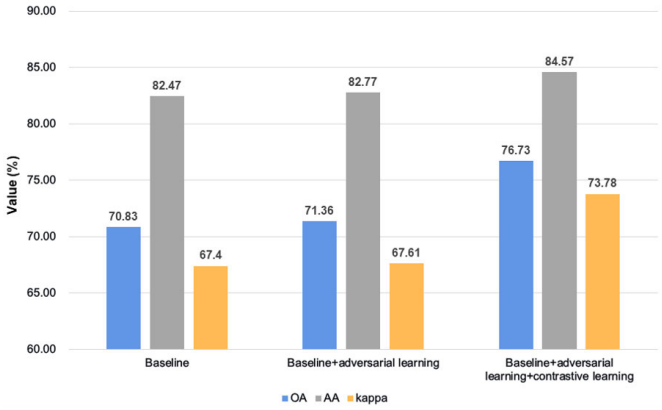



Fig. 17. Ablation experiment results on the IP dataset.

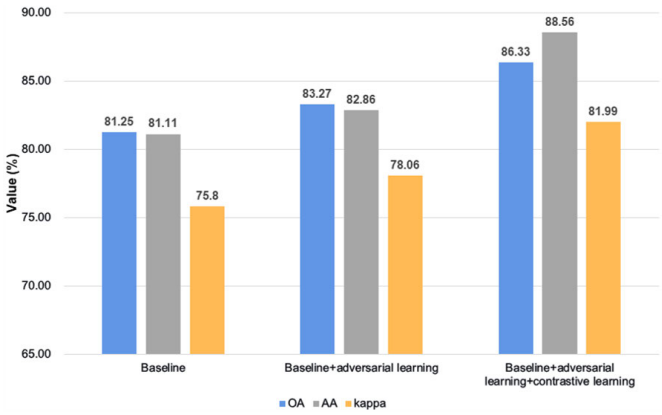



Fig. 18. Ablation experiment results on the UP dataset.

classifier that computes relationships is more suitable for this dataset than the linear classifier. The DCFSL approach mainly addresses the cross-domain problem, and it performs well when there are domain differences between the source and target domains. It usually utilizes domain-common features to reduce domain differences by means of domain alignment. The CDSTN method designs a new feature extractor for capturing features and implements domain alignment for features at different depths of the network. In addition, the operation of refining pseudolabels for samples enlarges the labeled dataset of the target domain, which further improves the performance of the model on the target domain classification task. The GMTN method aggregates graph structure, metalearning, and transfer learning to produce an integrated network that solves heterogeneous few-shot HSI classification tasks by enhancing feature learning capabilities. The neighborhood spatial information reset (NSIR) module enhances spatial and spectral information within the sample neighborhood, and the receptive field spectral attention (RFSA) mechanism reduces the impact of different categories of samples on discriminative spectral feature learning. In addition, learn more important information about the spatial structure of the graph based on the spectral feature similarity and distance consistency of surrounding pixels. The CTFSL method uses a new CT network to extract discriminative global and local features and reduces domain shifts using a domain discriminator. The excellent feature extraction capability of the model takes into account spatial details and global information, thus helping to improve performance on cross-domain tasks. Our method implements adversarial domain alignment on both the spectral alignment and semantic alignment levels. In addition, it also enhances the generalization of feature extractors to the target domain by contrastive learning.

*2) Ablation Experiments:* To demonstrate the usefulness of each of the proposed parts, we perform a set of ablation experiments on four datasets:

1) baseline;
2) baseline + adversarial learning;
3) baseline + adversarial learning + contrastive learning.

Among them, 1) serves as a baseline for other ablation experiments, which consists of a 3-D CNN that achieves few-shot classification by extracting domain-common features. To demonstrate the effect of the proposed adversarial learning strategy, 2) adds the losses associated with adversarial learning to 1), including the spectral alignment loss $L_{\text{local}}$ and the semantic alignment loss $L_{\text{global}}$. Since the process of computing the two losses involves the spectral alignment module and the semantic alignment module, this experiment can also show the contributions of the two modules on adversarial domain alignment; 3) demonstrates the effectiveness of contrastive learning strategy in improving our method. Similarly, the contrastive learning loss $L_{\text{cl}}$ generated by the contrastive learning module is added to this experiment.

Figs. 15–18 show the results of the ablation experiments. It can be seen that a baseline that uses only 3-D CNN

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TABLE X
RESULTS FOR DIFFERENT NUMBERS OF SAMPLES ON THE SALINAS DATASET (%)

| Method | $L=1$ | | | $L=2$ | | | $L=3$ | | | $L=4$ | | | $L=5$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa |
| SSRN | 61.93 ±4.22 | 66.25 ±3.96 | 58.83 ±4.61 | 67.24 ±2.08 | 73.04 ±1.46 | 63.30 ±1.93 | 78.65 ±2.26 | 80.41 ±2.36 | 75.27 ±1.79 | 84.05 ±2.85 | 85.25 ±1.02 | 81.00 ±1.50 | 86.21 ±2.53 | 93.02 ±2.13 | 82.65 ±1.28 |
| DFSL+SVM | 66.42 ±4.67 | 71.38 ±2.22 | 62.93 ±3.10 | 76.98 ±5.02 | 83.16 ±2.01 | 73.08 ±2.56 | 82.76 ±3.22 | 85.29 ±1.02 | 80.88 ±1.23 | 83.09 ±2.46 | 86.77 ±1.67 | 82.62 ±1.20 | 85.63 ±0.63 | 89.98 ±2.79 | 85.04 ±3.98 |
| DCFSL | 70.13 ±5.48 | 79.15 ±3.01 | 67.28 ±5.26 | 79.63 ±3.80 | 89.82 ±1.67 | 76.49 ±2.37 | 83.29 ±2.97 | 93.37 ±0.79 | 81.59 ±1.83 | 88.36 ±0.72 | 94.93 ±0.40 | 89.28 ±0.66 | 90.56 ±1.69 | 95.48 ±0.66 | 89.52 ±2.01 |
| Proposed | **77.25 ±3.65** | **86.35 ±2.18** | **74.83 ±2.62** | **88.10 ±2.52** | **90.62 ±1.73** | **86.69 ±1.80** | **89.96 ±1.31** | **94.30 ±0.86** | **88.78 ±1.00** | **91.61 ±1.28** | **95.66 ±0.53** | **90.02 ±0.91** | **92.80 ±1.22** | **96.71 ±0.50** | **92.65 ±0.43** |

TABLE XI
RESULTS FOR DIFFERENT NUMBERS OF SAMPLES ON THE PC DATASET (%)

| Method | $L=1$ | | | $L=2$ | | | $L=3$ | | | $L=4$ | | | $L=5$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa |
| SSRN | 68.04 ±5.66 | 59.22 ±5.01 | 64.83 ±3.42 | 79.62 ±2.02 | 75.33 ±0.98 | 77.56 ±1.21 | 85.43 ±1.30 | 82.10 ±0.76 | 83.21 ±0.58 | 93.11 ±0.89 | 90.05 ±0.12 | 92.02 ±0.47 | 96.72 ±0.76 | 91.67 ±0.38 | 94.84 ±1.02 |
| DFSL+SVM | 82.68 ±7.31 | 70.66 ±3.12 | 76.93 ±3.43 | 90.08 ±4.20 | 83.87 ±2.06 | 86.11 ±2.10 | 93.70 ±1.54 | 86.93 ±2.66 | 90.06 ±0.84 | 95.12 ±1.26 | 89.83 ±1.00 | 91.73 ±0.59 | 95.62 ±0.98 | 90.38 ±2.05 | 93.96 ±0.79 |
| DCFSL | 85.24 ±7.74 | 73.14 ±5.93 | 79.39 ±6.20 | 95.72 ±1.01 | 90.49 ±1.20 | 93.97 ±0.47 | 96.94 ±0.55 | 91.87 ±0.61 | 95.68 ±0.28 | 97.25 ±0.83 | 93.54 ±1.42 | 95.08 ±0.40 | 96.89 ±1.01 | 93.51 ±0.42 | 95.60 ±0.18 |
| Proposed | **93.69 ±5.08** | **89.49 ±3.88** | **91.15 ±3.62** | **96.95 ±1.21** | **93.24 ±1.01** | **95.70 ±0.92** | **97.24 ±0.96** | **92.09 ±0.40** | **96.09 ±0.33** | **97.43 ±0.84** | **93.61 ±0.38** | **96.39 ±0.43** | **97.98 ±1.22** | **95.32 ±0.70** | **97.26 ±0.30** |

TABLE XII
RESULTS FOR DIFFERENT NUMBERS OF SAMPLES ON THE IP DATASET (%)

| Method | $L=1$ | | | $L=2$ | | | $L=3$ | | | $L=4$ | | | $L=5$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa |
| SSRN | 31.67 ±6.62 | 40.06 ±4.53 | 28.44 ±6.88 | 44.62 ±4.30 | 48.04 ±2.01 | 40.20 ±2.16 | 50.99 ±2.18 | 54.06 ±1.58 | 46.43 ±1.33 | 61.74 ±1.22 | 62.87 ±0.36 | 54.88 ±0.40 | 62.76 ±0.35 | 64.12 ±0.03 | 58.45 ±0.21 |
| DFSL+SVM | 44.94 ±5.87 | 50.09 ±3.11 | 34.20 ±2.13 | 53.37 ±4.08 | 60.17 ±2.50 | 46.05 ±1.20 | 58.43 ±1.44 | 65.22 ±0.83 | 50.20 ±1.01 | 60.82 ±0.63 | 70.43 ±0.37 | 52.80 ±1.03 | 61.34 ±1.88 | 72.48 ±1.55 | 55.82 ±0.79 |
| DCFSL | 39.53 ±5.70 | 51.43 ±3.69 | 32.71 ±6.83 | 58.13 ±2.88 | 68.08 ±0.37 | 52.88 ±1.42 | 61.02 ±1.02 | 69.87 ±1.52 | 55.63 ±0.89 | 63.61 ±1.52 | 71.70 ±0.38 | 53.26 ±0.70 | 65.80 ±2.76 | 76.63 ±1.60 | 61.46 ±1.58 |
| Proposed | **50.59 ±4.22** | **60.10 ±3.43** | **44.87 ±4.75** | **59.21 ±2.32** | **68.40 ±2.26** | **55.36 ±3.17** | **66.07 ±1.30** | **77.99 ±0.83** | **61.87 ±1.66** | **74.56 ±1.58** | **85.78 ±0.50** | **71.57 ±1.22** | **76.73 ±1.40** | **84.57 ±0.54** | **73.78 ±1.44** |

TABLE XIII
RESULTS FOR DIFFERENT NUMBERS OF SAMPLES ON THE UP DATASET (%)

| Method | $L=1$ | | | $L=2$ | | | $L=3$ | | | $L=4$ | | | $L=5$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa | OA | AA | kappa |
| SSRN | 30.92 ±6.06 | 42.77 ±4.54 | 26.80 ±4.98 | 46.93 ±5.35 | 54.22 ±4.30 | 40.38 ±3.73 | 52.24 ±4.99 | 62.17 ±2.06 | 50.13 ±1.21 | 62.77 ±4.72 | 70.48 ±2.60 | 60.20 ±2.55 | 75.82 ±5.86 | 80.08 ±2.54 | 69.32 ±6.05 |
| DFSL+SVM | 45.49 ±8.30 | 56.58 ±3.79 | 35.99 ±2.89 | 58.24 ±8.04 | 62.30 ±2.42 | 48.22 ±3.37 | 64.37 ±3.65 | 68.85 ±1.88 | 60.11 ±1.20 | 70.65 ±4.01 | 72.04 ±2.58 | 66.80 ±2.62 | 79.19 ±1.11 | 75.82 ±1.23 | 73.10 ±1.52 |
| DCFSL | 46.53 ±8.19 | 63.64 ±5.40 | 37.54 ±6.78 | 60.19 ±9.22 | 71.39 ±3.80 | 51.13 ±4.62 | 69.63 ±5.26 | 75.71 ±2.63 | 62.30 ±3.30 | 78.26 ±3.96 | 79.59 ±2.06 | 71.94 ±2.17 | 84.49 ±1.18 | 83.71 ±1.77 | 79.81 ±1.60 |
| Proposed | **61.79 ±5.42** | **79.05 ±2.84** | **54.72 ±6.33** | **66.08 ±6.76** | **79.36 ±3.21** | **59.37 ±5.05** | **84.45 ±2.43** | **85.66 ±1.08** | **79.73 ±0.20** | **86.04 ±1.12** | **87.20 ±0.31** | **82.20 ±0.44** | **86.33 ±1.45** | **88.56 ±0.21** | **81.99 ±1.43** |

extracted features for few-shot classification has the lowest accuracy. The results show some improvement when adding an adversarial learning strategy and a contrastive learning strategy, respectively. For the Salinas dataset, the adversarial learning strategy and contrastive learning strategy improve 2.36% and 1.78% on OA, respectively. For the PC dataset, the adversarial learning strategy and contrastive learning strategy improve 0.16% and 0.53% on OA, respectively. For the IP dataset, the adversarial learning strategy and the contrastive learning strategy improve 0.53% and 5.37% on OA, respectively. For the UP dataset, the adversarial learning strategy and the contrastive learning strategy improve 2.02% and 3.06% on OA, respectively. Similar results are exhibited on the AA and kappa metrics.

Through the analysis, we can see that, when adding the losses associated with adversarial learning ($L_{\text{local}}$ and $L_{\text{global}}$) to the baseline, both the spectral alignment module and the semantic alignment module that we propose are also introduced. This enables the model to achieve further domain alignment at the spectral level and the high-level semantic level. When we add contrastive learning strategy to 2), the contrastive learning module produces contrastive learning loss $L_{\text{cl}}$. Therefore, 3) optimizes the few-shot classification task by improving the robustness and sensitivity of the model, which

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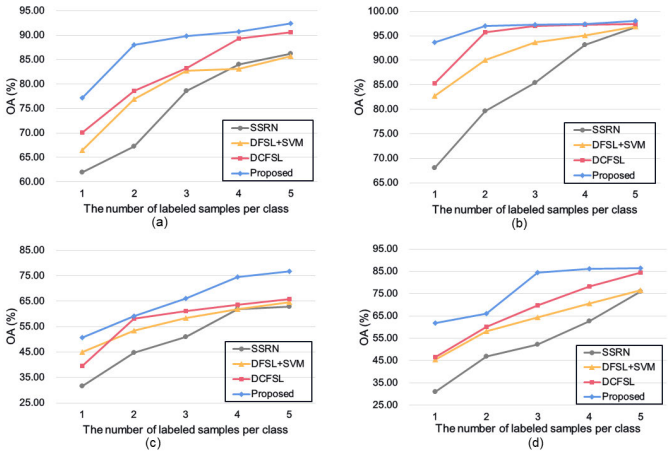



Fig. 19. Results for different numbers of labeled samples on four target datasets. (a) Salinas. (b) PC. (c) IP. (d) UP.

results in similar samples being classified closer together and dissimilar samples being classified farther apart. Therefore, the results of the ablation experiments can demonstrate the effectiveness of the adversarial learning and contrastive learning strategies that we introduce.

*3) Experiment About the Effect of Labeled Sample Size:* We perform experiments on Salinas, PC, IP, and UP datasets with different numbers of labeled samples from each class to verify the robustness of the proposed method. Moreover, our method is compared with SSRN, DCFL+SVM, and DCFSL. For each method, one, two, three, four, and five labeled samples are selected from each class, respectively. Tables X–XIII show the experimental results with different sample numbers for different methods. We use OA, AA, and kappa to evaluate. $L$ is the number of labeled samples per class in the target domain. For the Salinas dataset, the OA values of all methods increase as the number of labeled samples increases. The SSRN shows lower results because only joint spatial–spectral features are extracted for classification. Also, it can be found that the SSRN method results decrease more as the number of labeled samples decreases, thus leading to poor robustness and low accuracy. The DFSL + SVM method is improved over SSRN in terms of accuracy and robustness. The DCFSL method achieves better results than the previous two methods on the cross-domain problem but has low accuracy when the labeled samples are reduced. Compared with SSRN, DFSL + SVM, and DCFSL methods, our method not only obtains the highest accuracy but also outperforms other methods when the number of samples is reduced. Therefore, the proposed method has higher OA values than SSRN, DFSL + SVM, and DCFSL methods when $L$ adopts the same value. There are similar results on AA and kappa indicators. For the PC, IP, and UP datasets, we can obtain similar experimental results from Tables XI–XIII as for the Salinas dataset. The visualization results are shown in Fig. 19. It can be seen that our method still gives better results than other methods when the number of samples is small. This can prove that our method has robustness.

*4) Experiment About the Effect of Noise:* We perform experiments on four datasets, Salinas, PC, IP, and UP, respectively, and Tables XIV–XVII show the results of the OA metrics. For Gaussian and Poisson noises, the variance $\sigma^2$ of each noise is taken as 0.1, 1.0, and 5.0, respectively. Maximum values for different methods for each intensity are bolded. It can be seen that the result is not as good as the result without

TABLE XIV
EFFECT OF DIFFERENT NOISES ON THE SALINAS DATASET (%)

| Method | Gaussian | | | Poisson | | |
|---|---|---|---|---|---|---|
| | $\sigma^2=0.1$ | $\sigma^2=1.0$ | $\sigma^2=5.0$ | $\sigma^2=0.1$ | $\sigma^2=1.0$ | $\sigma^2=5.0$ |
| SVM | 77.49 ±0.63 | 74.10 ±0.26 | 68.77 ±0.59 | 75.08 ±2.62 | 73.05 ±1.16 | 66.89 ±1.30 |
| SSRN | 83.50 ±0.15 | 80.32 ±0.07 | 76.95 ±0.10 | 83.20 ±0.14 | 80.04 ±0.21 | 76.05 ±0.06 |
| DFSL | 83.51 ±0.22 | 80.46 ±0.21 | 76.82 ±0.08 | 83.17 ±0.32 | 80.25 ±0.17 | 76.21 ±0.13 |
| RN-FSC | 83.03 ±0.05 | 80.11 ±0.32 | 76.04 ±0.61 | 80.22 ±0.85 | 77.18 ±0.33 | 74.40 ±0.51 |
| DCFSL | 87.16 ±0.83 | 86.72 ±0.11 | 85.40 ±0.14 | 86.46 ±0.31 | 84.75 ±0.57 | 82.65 ±0.42 |
| CDSTN | 88.05 ±0.72 | 86.53 ±0.26 | 85.87 ±0.64 | 87.20 ±0.36 | 85.17 ±0.40 | 83.06 ±0.28 |
| GMTN | 90.06 ±0.04 | 87.70 ±0.33 | 86.21 ±0.17 | 88.13 ±0.84 | 85.63 ±0.77 | 83.82 ±0.35 |
| CTFSL | 91.05 ±0.78 | 88.26 ±0.54 | 86.82 ±0.50 | 88.90 ±1.03 | 86.76 ±1.20 | 84.44 ±0.61 |
| Proposed | **91.75 ±1.02** | **89.69 ±0.20** | **90.05 ±0.11** | **89.45 ±1.21** | **89.44 ±1.15** | **87.90 ±0.26** |

TABLE XV
EFFECT OF DIFFERENT NOISES ON THE PC DATASET (%)

| Method | Gaussian | | | Poisson | | |
|---|---|---|---|---|---|---|
| | $\sigma^2=0.1$ | $\sigma^2=1.0$ | $\sigma^2=5.0$ | $\sigma^2=0.1$ | $\sigma^2=1.0$ | $\sigma^2=5.0$ |
| SVM | 89.72 ±0.57 | 86.11 ±0.74 | 82.32 ±0.62 | 89.22 ±0.24 | 86.07 ±0.27 | 81.90 ±0.84 |
| SSRN | 93.12 ±0.52 | 90.04 ±0.62 | 88.52 ±0.21 | 93.05 ±0.02 | 89.95 ±0.06 | 87.26 ±0.83 |
| DFSL | 94.02 ±0.45 | 91.68 ±0.34 | 89.96 ±0.75 | 93.89 ±0.56 | 91.47 ±0.73 | 88.73 ±0.43 |
| RN-FSC | 94.55 ±0.46 | 93.20 ±0.75 | 92.05 ±0.23 | 94.02 ±0.21 | 93.10 ±0.14 | 91.73 ±0.15 |
| DCFSL | 97.16 ±0.05 | 94.90 ±1.05 | 92.91 ±0.88 | 96.83 ±0.56 | 95.26 ±0.31 | 92.28 ±0.46 |
| CDSTN | 97.24 ±0.21 | 95.63 ±0.53 | 94.21 ±0.35 | 96.91 ±0.13 | 95.74 ±0.52 | 94.00 ±0.21 |
| GMTN | 97.20 ±0.35 | 96.82 ±0.42 | 95.11 ±0.46 | 97.01 ±0.45 | 96.36 ±0.32 | 94.87 ±0.42 |
| CTFSL | 97.38 ±0.13 | 96.88 ±0.37 | 95.42 ±0.43 | 97.20 ±0.34 | 96.33 ±0.15 | 95.21 ±0.46 |
| Proposed | **97.74 ±0.25** | **97.36 ±0.41** | **96.95 ±0.39** | **97.56 ±0.47** | **96.62 ±0.60** | **96.42 ±0.37** |

TABLE XVI
EFFECT OF DIFFERENT NOISES ON THE IP DATASET (%)

| Method | Gaussian | | | Poisson | | |
|---|---|---|---|---|---|---|
| | $\sigma^2=0.1$ | $\sigma^2=1.0$ | $\sigma^2=5.0$ | $\sigma^2=0.1$ | $\sigma^2=1.0$ | $\sigma^2=5.0$ |
| SVM | 38.57 ±0.76 | 34.66 ±0.35 | 29.75 ±1.45 | 37.36 ±2.53 | 35.48 ±0.66 | 28.57 ±0.38 |
| SSRN | 57.72 ±0.33 | 55.68 ±0.54 | 51.03 ±0.88 | 56.24 ±0.98 | 52.89 ±0.36 | 48.55 ±0.72 |
| DFSL | 57.22 ±0.46 | 55.73 ±0.24 | 52.05 ±0.83 | 57.02 ±0.20 | 54.80 ±0.89 | 50.73 ±0.29 |
| RN-FSC | 55.34 ±0.74 | 53.11 ±0.21 | 49.06 ±0.30 | 54.22 ±0.15 | 51.76 ±0.42 | 48.09 ±0.23 |
| DCFSL | 65.53 ±0.94 | 62.46 ±1.20 | 60.48 ±1.03 | 65.04 ±0.77 | 54.34 ±2.62 | 47.79 ±2.95 |
| CDSTN | 66.03 ±0.62 | 63.63 ±0.24 | 60.72 ±0.75 | 65.25 ±0.78 | 61.46 ±1.36 | 57.40 ±0.02 |
| GMTN | 66.64 ±0.67 | 63.30 ±0.65 | 60.85 ±0.36 | 66.24 ±0.35 | 60.75 ±0.23 | 56.22 ±0.63 |
| CTFSL | 65.83 ±1.20 | 65.20 ±0.74 | 61.07 ±0.64 | 65.56 ±0.37 | 63.13 ±0.19 | 58.85 ±0.56 |
| Proposed | **73.71 ±1.61** | **74.34 ±2.13** | **73.58 ±2.90** | **77.38 ±0.96** | **67.73 ±0.46** | **65.91 ±2.51** |

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TABLE XVII
EFFECT OF DIFFERENT NOISES ON THE UP DATASET (%)

| Method | Gaussian | | | Poisson | | |
|---|---|---|---|---|---|---|
| | $\sigma^2 = 0.1$ | $\sigma^2 = 1.0$ | $\sigma^2 = 5.0$ | $\sigma^2 = 0.1$ | $\sigma^2 = 1.0$ | $\sigma^2 = 5.0$ |
| SVM | 58.34 ±0.41 | 52.76 ±0.28 | 46.29 ±0.08 | 56.72 ±0.96 | 51.04 ±0.02 | 43.63 ±1.02 |
| SSRN | 71.58 ±0.23 | 65.06 ±0.75 | 59.20 ±1.26 | 69.37 ±1.40 | 62.46 ±1.02 | 58.65 ±0.24 |
| DFSL | 75.05 ±0.14 | 69.31 ±0.31 | 62.27 ±0.08 | 74.20 ±0.13 | 68.26 ±0.22 | 60.83 ±0.49 |
| RN-FSC | 77.26 ±0.06 | 70.43 ±0.78 | 63.50 ±0.53 | 76.42 ±0.34 | 70.08 ±0.63 | 62.21 ±0.12 |
| DCFSL | 79.41 ±2.54 | 73.02 ±3.60 | 67.31 ±1.12 | 75.67 ±2.41 | 68.80 ±1.83 | 63.08 ±3.93 |
| CDSTN | 81.20 ±0.12 | 77.47 ±0.36 | 71.62 ±0.43 | 80.10 ±0.87 | 75.65 ±0.82 | 69.04 ±1.31 |
| GMTN | 82.32 ±0.83 | 78.82 ±0.26 | 74.20 ±0.15 | 80.35 ±0.22 | 76.34 ±0.75 | 71.73 ±0.03 |
| CTFSL | 84.64 ±2.10 | 80.22 ±0.54 | 77.75 ±0.84 | 82.02 ±0.64 | 79.32 ±0.23 | 75.74 ±0.37 |
| Proposed | **86.67 ±3.33** | **83.39 ±1.06** | **79.18 ±0.42** | **83.96 ±5.60** | **80.81 ±2.47** | **77.93 ±1.83** |

adding noise when the noise is added. Moreover, we can analyze different intensities of the same noise for a single dataset. For example, for the Salinas dataset, when Gaussian noise is used, the accuracy is higher for $\sigma^2 = 0.1$ (91.75%) than for $\sigma^2 = 1.0$ (89.69%) and $\sigma^2 = 5.0$ (90.05%). The same result is observed when Poisson noise is used. Similarly, we discuss the effects of different noises of the same intensity. When $\sigma^2 = 0.1$, comparing Gaussian and Poisson noises, it can be seen that Gaussian noise (91.75%) has better accuracy than Poisson noise (89.45%) on the Salinas dataset. The same result is obtained for $\sigma^2 = 1.0$ and $\sigma^2 = 5.0$. In addition, it can be seen that our method performs better than other methods on results with the same intensity or the same type of noise. The results are roughly similar to the rest of the dataset.

By analyzing the results, it can be seen that the classification results are affected by the type and intensity of the noise. For the same type of noise, when $\sigma^2$ is smaller, the model is more likely to learn features from the data, which makes it more beneficial to achieve the cross-domain classification task. Therefore, as $\sigma^2$ increases, the generalization of the model decreases overall. Of course, it is not excluded that there are cases where the results are better when $\sigma^2$ is larger due to the dataset or other factors. For different noises of the same intensity, Gaussian noise gives better results than Poisson noise. It also shows that our model is more robust in the face of Gaussian noise than Poisson noise. Moreover, our method keeps the model more robust than other compared methods when noise is present. Thus, the superiority of our method can be reflected.

## V. CONCLUSION

We propose an adversarial domain alignment with a contrastive learning method for the task of HSI classification. Domain alignment is implemented as an adversarial learning method with spectral and semantic alignments for transferring knowledge at the local and global levels, respectively. In semantic alignment, the MSS module extracts different scales of object information for different domains. In addition, contrastive learning enhances the representation of source and target domain features in a self-supervised manner, which acquires more information about the target domain and enhances the robustness and sensitivity of the feature extractor. We demonstrate the effectiveness of the proposed method by comparing it with other methods through a series of experiments. Thus, our method has a good performance in HSI classification and can handle small sample problems well, while the task of HSI classification is still limited by some inherent characteristics, such as "low spatial resolution," "small labeled samples," "the same objects with different spectrum," and "different objects with the same spectrum." Fortunately, some other data resources, including light detection and ranging (LiDAR) and SAR, exhibit complementary information to HSI. Therefore, cross-modal learning and classification would be attractive and have been proven to be effective by some initial research. In the future, we would like to explore the domain-gap between different data modals (i.e., HSI and SAR), which benefits the classification performance and the comprehensive utilization of various remote sensing data.

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